5G RAN

# Remote Interference Management (Low-Frequency TDD) Feature Parameter Description


**Issue** Draft B

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# Contents

# 1 Change History

The following describes the specific changes in different issues.

## 5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added the impact relationship between GP symbol shutdown in special slots and proactive avoidance of remote interference. For details, see **6.3.2.3 Function Impacts**. | None | Low-frequency TDD | 3900 and 5900 series base stations (macro base stations) |

- Editorial changes

  Added the PUSCH scheduling prohibition in self-contained slots function. For details, see **6.2.2 Impacts**.

  Revised descriptions of the impact relationships of the "proactive avoidance of remote interference" function with "uplink intra-band CA" and "uplink intra-FR inter-band CA". For details, see **6.3.2.3 Function Impacts**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

## 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-12-14).

- Technical changes

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added the impact relationship between "RedCap Introduction" and "UL and DL Decoupling" in scenarios where "proactive avoidance of remote interference" is enabled. For details, see **6.3.2.3 Function Impacts**. | None | Low-frequency TDD | 3900 and 5900 series base stations (macro base stations) |
| Added the mutually exclusive relationship among the following functions: "proactive avoidance of remote interference", "positioning-dedicated SRS transmission in uplink slots", and "SRS and PUSCH/PUCCH concurrency compatibility". For details, see **6.3.2.2 Mutually Exclusive Functions**. | None | Low-frequency TDD | 3900 and 5900 series base stations (macro base stations) |

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added the impact relationships of remote interference adaptive avoidance and enhanced remote interference adaptive avoidance with uplink symbol power saving optimization. For details, see **Remote Interference Adaptive Avoidance** and **Enhanced Remote Interference Adaptive Avoidance**. | None | Low-frequency TDD | 3900 and 5900 series base stations (macro base stations) |
| Removed the mutually exclusive relationships of proactive avoidance of remote interference with UL and DL Decoupling and NUL 1Tx to SUL 1Tx Switching Time. For details, see **6.3.2.2 Mutually Exclusive Functions**. | None | Low-frequency TDD | 3900 and 5900 series base stations (macro base stations) |

- Editorial changes

  Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

# 2 About This Document

## 2.1 General Statements

### Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

**NOTE**

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

### Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

## 2.2 Features in This Document

This document describes the following features.

| Feature ID | Feature Name | Chapter/Section |
| --- | --- | --- |
| FOFD-030208 | Remote Interference Management (RIM) | 4 Remote Interference Management |
| FOFD-061220 | Adaptive Interference Suppression | 5.1 Remote Interference Adaptive Suppression<br>5.2 Remote Interference Resistance for PUSCH Demodulation<br>5.3 Remote Interference Resistance for Random Access |
| FOFD-061292 | Proactive Avoidance of Remote Interference | 6 Proactive Avoidance of Remote Interference |

## 2.3 Differences

### Differences Between NSA and SA

| Function Name | Difference | Chapter/Section |
| --- | --- | --- |
| Remote interference management | None | 4 Remote Interference Management |
| Remote interference adaptive suppression | None | 5.1 Remote Interference Adaptive Suppression |
| Remote interference resistance for PUSCH demodulation | None | 5.2 Remote Interference Resistance for PUSCH Demodulation |
| Remote interference resistance for random access | None | 5.3 Remote Interference Resistance for Random Access |
| Proactive avoidance of remote interference | None | 6 Proactive Avoidance of Remote Interference |
| Symbol configuration for channel calibration | None | 7 Symbol Configuration for Channel Calibration |

### Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

| Function Name | Difference | Chapter/Section |
|---|---|---|
| Remote interference management | Supported only in low frequency bands | **4 Remote Interference Management** |
| Remote interference adaptive suppression | Supported only in low frequency bands | **5.1 Remote Interference Adaptive Suppression** |
| Remote interference resistance for PUSCH demodulation | Supported only in low frequency bands | **5.2 Remote Interference Resistance for PUSCH Demodulation** |
| Remote interference resistance for random access | Supported only in low frequency bands | **5.3 Remote Interference Resistance for Random Access** |
| Proactive avoidance of remote interference | Supported only in low frequency bands | **6 Proactive Avoidance of Remote Interference** |
| Symbol configuration for channel calibration | Supported only in low frequency bands | **7 Symbol Configuration for Channel Calibration** |

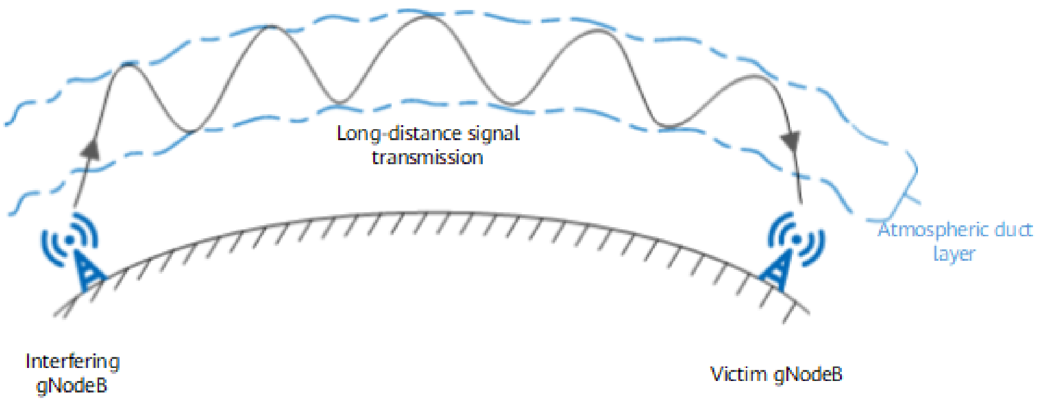
# 3 Overview

## Generation of Remote Interference

In this document, remote interference refers to the interference that occurs in a TDD system as a result of atmospheric ducting.

Atmospheric ducting, as illustrated in **Figure 3-1**, is a type of phenomenon that is observed in the troposphere. An atmospheric duct can be formed in the troposphere due to temperature inversion or a rapid decrease in water vapor with increased altitude. Such a duct acts as an atmospheric waveguide, trapping electromagnetic waves that propagate within it. This phenomenon is known as atmospheric ducting and is similar to propagation of electromagnetic waves within a metal waveguide. Atmospheric ducting has two main impacts on electromagnetic waves. First, it increases the electric field strength, and second, it extends propagation distances. Therefore, electromagnetic waves that are trapped by an atmospheric duct can travel large distances.

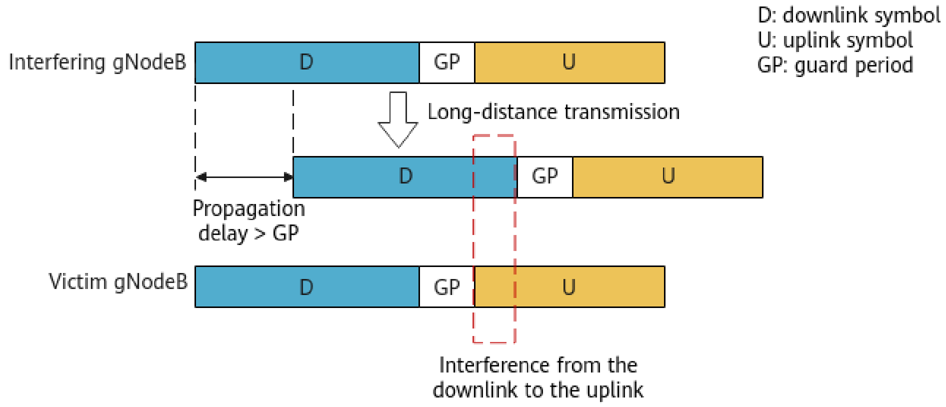
**Figure 3-1** Atmospheric ducting




A guard period (GP) is defined in the slot structure of a TDD system to separate uplink slots from downlink slots, thereby avoiding mutual interference between uplink and downlink signals. Due to atmospheric ducting, **signals from a gNodeB travel a large distance and interfere with a distant gNodeB. If the propagation delay exceeds the GP, downlink signals from an interfering gNodeB can be received by a victim gNodeB in uplink slots. The downlink transmissions can therefore interfere with uplink reception at the victim**

**gNodeB. Such interference is referred to as remote intra-frequency interference** (see Figure 3-2). Due to reciprocity in atmospheric ducting, downlink signals from the victim gNodeB also interfere with uplink reception at the interfering gNodeB. As a result, KPIs of the entire network deteriorate.

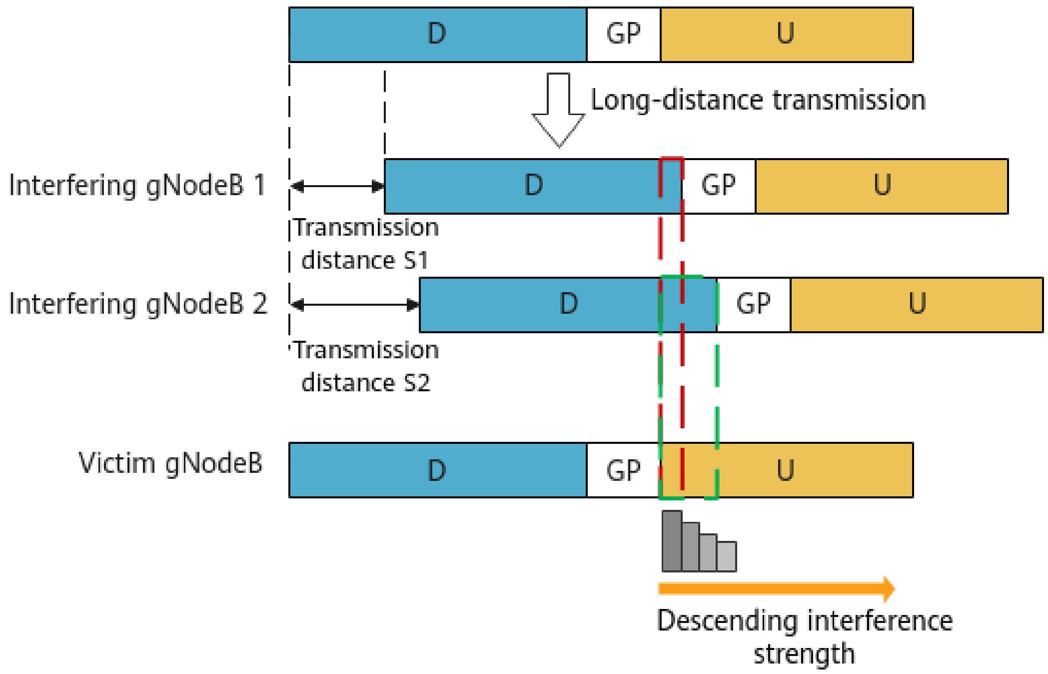
**Figure 3-2** Generation of remote interference




## Sloping Character of Remote Interference

Remote interference exhibits a sloping character, meaning that **in a victim cell, the interference is stronger at the uplink symbol closer to the GP**. This is because the interference is caused by accumulated signals from multiple sources with different signal transmission distances. The sloping character can be used to identify remote interference at a cell. Figure 3-3 illustrates the sloping character of remote interference.

**Figure 3-3** Sloping character of remote interference



## Remote Interference Management Solution

To solve remote interference problems caused by atmospheric ducting, the remote interference management solution is introduced. **Table 3-1** describes the functions included in this solution.

**Table 3-1** Remote interference management solution

| Category | Feature | Function Name | Principles | Reference |
|---|---|---|---|---|
| Active avoidance on the interference source side | FOFD-030208 Remote Interference Management (RIM) | Remote interference adaptive avoidance | After detecting remote interference management reference signals (RIM-RSs), the interfering cell increases its GP to reduce remote interference to other cells. | **4 Remote Interference Management** |
| | | Enhanced remote interference adaptive avoidance | After detecting RIM-RSs, the interfering cell adjusts the downlink scheduling positions of UEs with high interference to reduce remote interference to other cells. | **4.1.3 Enhanced Remote Interference Adaptive Avoidance** |
| Passive avoidance on the victim side | FOFD-061220 Adaptive Interference Suppression | Remote interference adaptive suppression | After detecting remote interference, a victim cell adjusts the MCS index for uplink scheduling in the victim symbols to reduce the impact of remote interference. | **5.1 Remote Interference Adaptive Suppression** |
| | | Remote interference resistance for PUSCH demodulation | The channel estimation result is adjusted to enhance PUSCH demodulation performance. | **5.2 Remote Interference Resistance for PUSCH Demodulation** |
| | | Remote interference resistance for random access | The related thresholds and scheduling parameters in the access phase are adaptively adjusted to increase the random access success rate. | **5.3 Remote Interference Resistance for Random Access** |

| Category | Feature | Function Name | Principles | Reference |
| --- | --- | --- | --- | --- |
| Passive avoidance on the victim side | FOFD-061292 Proactive Avoidance of Remote Interference | Proactive avoidance of remote interference | SRSs are arranged at time-domain positions with low uplink interference, implementing proactive and flexible interference avoidance. | **6 Proactive Avoidance of Remote Interference** |
| Passive avoidance on the victim side | None | Symbol configuration for channel calibration | This function enables channel calibration sequences to be transmitted in symbols suffering from weak remote interference. This mitigates the impacts of remote interference on channel calibration, thereby increasing the channel calibration success rate. | **7 Symbol Configuration for Channel Calibration** |

# 4 Remote Interference Management

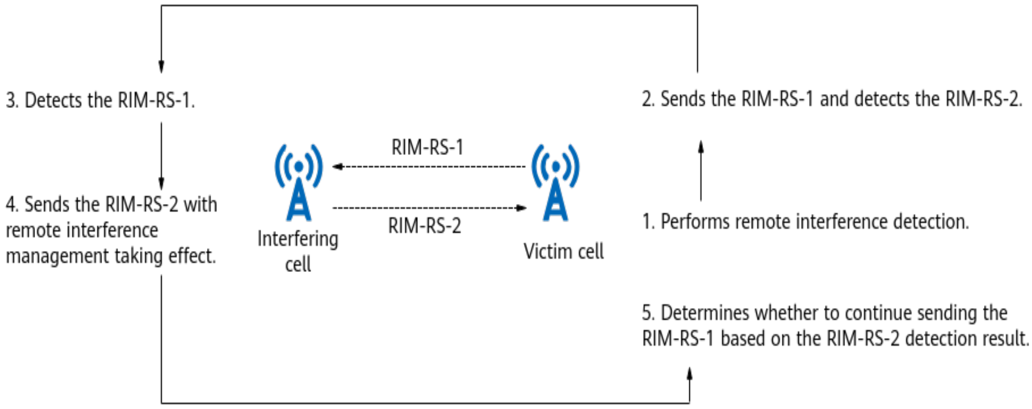
## 4.1 Principles

The Remote Interference Management feature includes the following functions: remote interference adaptive avoidance and enhanced remote interference adaptive avoidance. **Table 4-1** describes the switches used for controlling the functions. The working process shown in **Figure 4-1** starts when any of the functions is enabled, but the whole process can take effect only when the corresponding switch is turned on in both the victim and interfering cells.

**Table 4-1** Functions of remote interference management

| Function Name | Function Switch | Reference |
|---|---|---|
| Remote interference adaptive avoidance | **RIM_ADAPT_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter | **4.1.2 Remote Interference Adaptive Avoidance** |
| Enhanced remote interference adaptive avoidance | **RIM_ADAPT_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter | **4.1.3 Enhanced Remote Interference Adaptive Avoidance** |

**Figure 4-1** Working process of remote interference management




1. Performs remote interference detection.

   When the gNodeB serving a victim cell detects interference in uplink symbols, the gNodeB sends the RIM-RS-1 if the interference strength in an uplink symbol decreases as the distance between the uplink symbol and the GP increases (meeting the slope characteristics of the atmospheric duct) and the power difference between uplink symbols is greater than the value of the **NRDUCellRimConfig**.***UlPwrDiffThld*** parameter.

2. Sends the RIM-RS-1 and detects the RIM-RS-2.

   The victim cell periodically sends the RIM-RS-1 in the last two downlink symbols of the self-contained slot and starts RIM-RS-2 detection. For details about the RIM-RS, see **4.1.1 RIM-RS**.

3. Detects the RIM-RS-1.

   Due to the reciprocity of atmospheric ducts, the interfering cell also experiences remote interference from the peer end. The gNodeB serving the interfering cell periodically detects the RIM-RS-1 in uplink symbols to measure the strength of remote interference to the cell and identify the symbol position of the peer end, as shown in **Figure 4-2**. Whether to validate or exit remote interference management depends on the RIM-RS-1 detection result.

   – The conditions for validating remote interference management are as follows:

     Remote interference adaptive avoidance or enhanced remote interference adaptive avoidance takes effect if the gNodeB detects the RIM-RS-1 and the RIM-RS-1 signal strength meets the following conditions within a number of consecutive periods (specified by the **NRDUCellRimConfig**.***IntrfAvoidDetectPrdNum*** parameter):

     ■ Near-far functionality for type-1 RIM-RS is disabled: The RIM-RS-1 signal strength is greater than the value of the **NRDUCellRimConfig**.***RimRsType1IntrfThld*** parameter.

     ■ Near-far functionality for type-1 RIM-RS is enabled: The RIM-RS-1 signal strength is greater than the sum of the **NRDUCellRimConfig**.***RimRsType1IntrfThld*** and **NRDUCellRimConfig**.***RimRsType1IntrfThldOfs*** parameter values.

     For details about near-far functionality for type-1 RIM-RS, see **Near-Far Functionality for Type-1/Type-2 RIM-RS**.

– The conditions for exiting remote interference management are as follows:

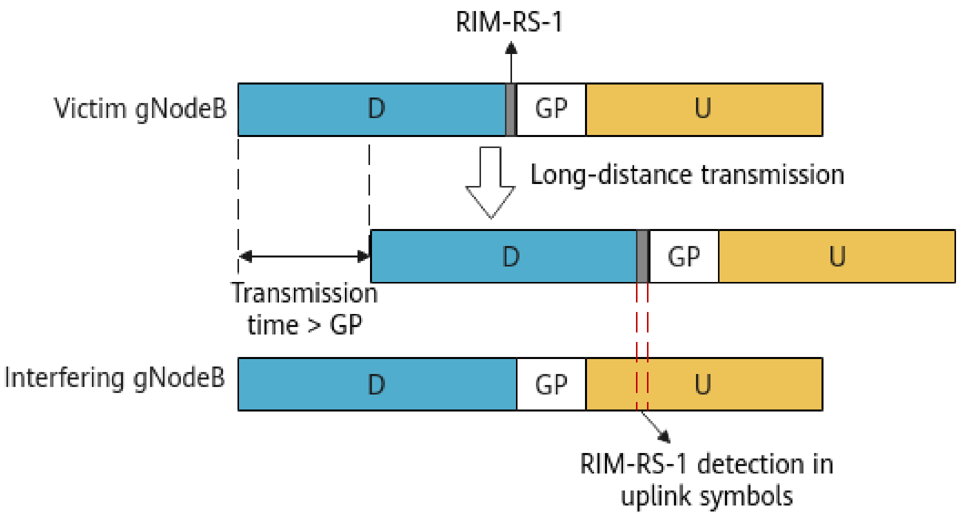
  After remote interference management takes effect, if the gNodeB does not detect the RIM-RS-1 or the RIM-RS-1 signal strength does not meet the requirements within a number of consecutive periods (specified by **NRDUCellRimConfig**.***IntrfAvoidRstrDetectPrdNum***), remote interference management is terminated, and the gNodeB restores to the state before remote interference management takes effect.

NOTE

When **NRDUCellRimConfig**.***IntrfAvoidDetectPrdNum*** is set to **0**:

- If the power difference between uplink symbols is greater than the **NRDUCellRimConfig**.***IntrfAvoidUlPwrDiffThld*** parameter value, remote interference adaptive avoidance directly takes effect for gNodeBs without depending on the RIM-RS detection result.
- The default value **3** of this parameter takes effect for enhanced remote interference adaptive avoidance.

**Figure 4-2** RIM-RS-1 remote interference detection




4. Sends the RIM-RS-2 with remote interference management taking effect.

   Remote interference adaptive avoidance or enhanced remote interference adaptive avoidance takes effect in the interfering cell based on the function switch settings. For details, see **4.1.2 Remote Interference Adaptive Avoidance** and **4.1.3 Enhanced Remote Interference Adaptive Avoidance**. At the same time, the interfering cell sends the RIM-RS-2 in the last two downlink symbols of the self-contained slot.

5. Determines whether to continue sending the RIM-RS-1 based on the RIM-RS-2 detection result.

   The victim cell continues sending the RIM-RS-1 when detecting the RIM-RS-2 in uplink symbols. Otherwise, it stops sending the RIM-RS-1.

NOTE

According to the reciprocity of atmospheric ducts, victim cells can also be interfering cells that cause remote interference to other cells, and vice versa. In all such cases, remote interference management works in the same way.

### 4.1.1 RIM-RS

The RIM-RS is a sequence of 15 bits to 22 bits and sent over downlink symbols. It is used for remote interference detection. As shown in **Figure 4-1**, RIM-RSs are classified into two types: RIM-RS-1 and RIM-RS-2. To support mutual RIM-RS check between devices of different vendors, RIM-RS-1 and RIM-RS-2 must support different configurations. As such, Huawei introduces the type-2 RIM-RS function.

Type-2 RIM-RS is controlled by the **RIM_RS2_SW** option of the **NRDUCellRimConfig**.***RimRsFunctionSwitch*** parameter. This function affects the RIM-RS-1 and RIM-RS-2 configurations, as described in **Table 4-2**.

**Table 4-2** RIM-RS-1 and RIM-RS-2 configurations

| Status of the RIM_RS2_SW Option | RIM-RS-1 | RIM-RS-2 |
| --- | --- | --- |
| Deselected | Type-1 RIM-RS configuration | Type-1 RIM-RS configuration |
| Selected | Type-1 RIM-RS configuration | Type-2 RIM-RS configuration |

For details about the configuration items related to type-1 RIM-RS and type-2 RIM-RS, see **Type-1 RIM-RS** and **Type-2 RIM-RS**, respectively.

The near-far functionality for type-1 RIM-RS and near-far functionality for type-2 RIM-RS are introduced to extend the detection distance of remote interference sources from about 200 km to about 400 km by increasing the RIM-RS transmission occasions. For details, see **Near-Far Functionality for Type-1/Type-2 RIM-RS**.

#### Type-1 RIM-RS

The gNodeB generates an RIM-RS based on the following protocol-defined parameters. For details, see 3GPP TS 38.211 V16.5.0 and 3GPP TS 28.541 V16.1.0.

**Table 4-3** Type-1 RIM-RS configuration parameters

| Parameter Name | Parameter ID | Function |
| --- | --- | --- |
| RIM-RS Type1 Set ID Size | **NRDUCellRimConfig**.***RimRsType1SetIdSize*** | This parameter determines the maximum number of interference sources that can be detected. |
| RIM-RS Type1 Set ID | **NRDUCellRimConfig**.***RimRsType1SetId*** | This parameter uniquely identifies an interference source. |

| Parameter Name | Parameter ID | Function |
| --- | --- | --- |
| RIM-RS Frequency Candidate Number | **NRDUCellRimConfig**.***RimRsFreqCandNum*** | This parameter indicates the number of candidate frequency domain resources for type-1 RIM-RS. If the interfering and victim cells differ in bandwidth (bandwidth size or frequency-domain position), the number of candidate frequency domain resources for RIM-RS is configured based on the overlapped bandwidth. |
| RIM-RS Type1 Sequence Candidate Number | **NRDUCellRimConfig**.***RimRsType1SeqCandNum*** | This parameter indicates the number of type-1 RIM-RSs that can be transmitted on a frequency domain resource. |
| RIM-RS Type1 Scrambling ID for IS0 to RIM-RS Type1 Scrambling ID for IS7 | **NRDUCellRimConfig**.***RimRsType1ScidForIs0*** to **NRDUCellRimConfig**.***RimRsType1ScidForIs7*** | This parameter uniquely identifies a RIM-RS. If *n* RIM-RS candidate resources are configured, set the first *n* scrambling IDs to different effective values (0 to 1023). |
| RIM-RS Scramble Timer Multiplier Factor | **NRDUCellRimConfig**.***RimRsScrTimerMultiFactor*** | The parameters are used to specify the scrambling timer factor. They together with the type-1 RIM-RS scrambling ID determine a RIM-RS. |
| RIM-RS Scramble Timer Offset | **NRDUCellRimConfig**.***RimRsScrTimerOffset*** | |

To ensure correct RIM-RS detection, the RIM-RS start frequency-domain positions must be completely the same for the cell that sends the RIM-RS (referred to as sending cell) and the cell for RIM-RS detection (referred to as detecting cell). Given this, Huawei introduces the following two frequency-domain offset parameters to determine the RIM-RS start frequency-domain position of a cell:

- **NRDUCellRimConfig**.***RimRsStartFreqOffset***: It specifies the offset (in units of RB) of the start frequency-domain for RIM-RS. The reference point is Point A.
- **NRDUCellRimConfig**.***RimRsStartFreqReOffset***: It specifies the offset (in units of RE) of the start frequency-domain for RIM-RS. The reference point is Point A plus the value of the **NRDUCellRimConfig**.***RimRsStartFreqOffset*** parameter.

The RIM-RS start frequency-domain position of a cell is calculated using the following formula:

RIM-RS start frequency-domain position of a cell = Point A of the cell + **NRDUCellRimConfig**.***RimRsStartFreqOffset*** + **NRDUCellRimConfig**.***RimRsStartFreqReOffset***

If the bandwidths of the sending cell and the detecting cell are completely aligned, default values **0** are used for the **NRDUCellRimConfig**.***RimRsStartFreqOffset*** and **NRDUCellRimConfig**.***RimRsStartFreqReOffset*** parameters of both cells. In this case, the RIM-RS start frequency-domain positions of the cells are aligned with their Point A.

If the bandwidths of the sending cell and the detecting cell are not aligned, the frequency-domain offset parameters need to be configured to adjust the cells' RIM-RS start frequency-domain positions to be the same. The following describes how to set the parameters in two scenarios.

- Scenario 1:

  As shown in **Figure 4-3**, cells A and B belong to different operators, and their frequency-domain information is listed in **Table 4-4**. With Point A of cell B as the reference, the **NRDUCellRimConfig**.***RimRsStartFreqOffset*** and **NRDUCellRimConfig**.***RimRsStartFreqReOffset*** parameters must be set to **0** for cell B. The difference between Point A of the two cells is 3 REs. Therefore, the **NRDUCellRimConfig**.***RimRsStartFreqOffset*** and **NRDUCellRimConfig**.***RimRsStartFreqReOffset*** parameters must be set to **0** and **3** respectively for cell A.

  **Figure 4-3** RIM-RS frequency-domain positions when cell bandwidths are different

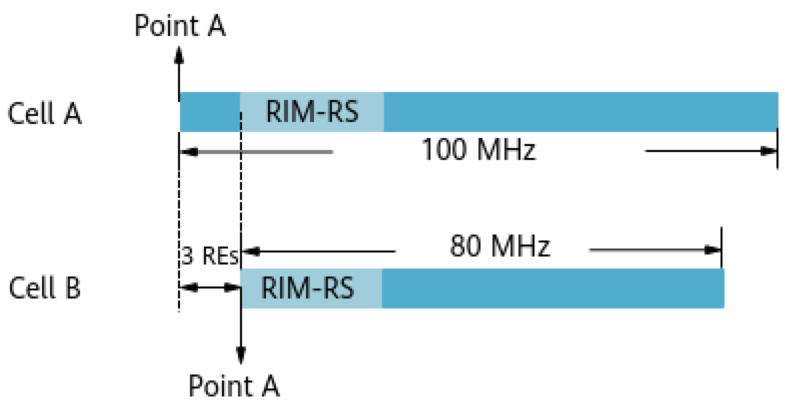


**Table 4-4** Frequency-domain information of cells with different bandwidths

| Cell | Frequency Band | Cell Center Frequency Number | Frequency at Point A | Subcarrier Spacing (SCS) | Frequency Difference | Number of RBs and REs Corresponding to the Frequency Difference |
|---|---|---|---|---|---|---|
| A | 3500–3600 MHz | 636666 | 3500.85 MHz | 30 kHz | 90 kHz | 0 RB+3 REs |
| B | 3500–3580 MHz | 636000 | 3500.94 MHz | 30 kHz | 0 (cell B being used as a reference) | 0 RB+0 RE |

- Scenario 2:

  As shown in **Figure 4-4**, the bandwidths of cell A and cell B partially overlap. **Table 4-5** describes the frequency-domain information of the cells. With Point A of cell B as the reference, the **NRDUCellRimConfig**.***RimRsStartFreqOffset*** and **NRDUCellRimConfig**.***RimRsStartFreqReOffset*** parameters must be set to **0** for cell B. The difference between Point A of the two cells is 166 RBs plus 8 REs. Therefore, the **NRDUCellRimConfig**.***RimRsStartFreqOffset*** and **NRDUCellRimConfig**.***RimRsStartFreqReOffset*** parameters must be set to **166** and **8** respectively for cell A.

**Figure 4-4** RIM-RS frequency-domain positions when cell bandwidths overlap

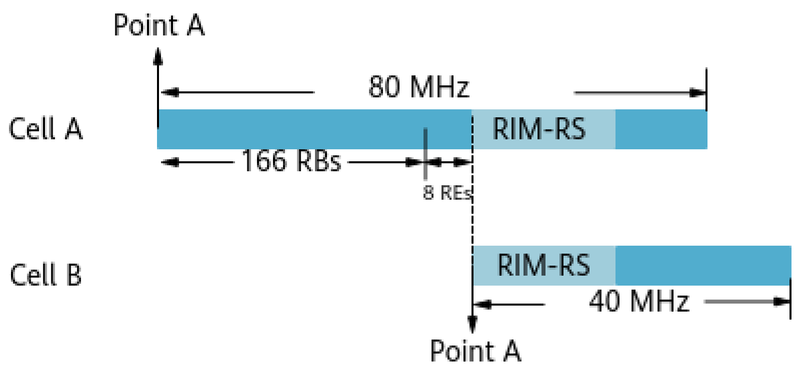


**Table 4-5** Frequency-domain information of cells with bandwidth overlap

| Cell | Frequency Band | Cell Center Frequency Number | Frequency at Point A | SCS | Frequency Difference | Number of RBs and REs Corresponding to the Frequency Difference |
|---|---|---|---|---|---|---|
| A | 3500–3580 MHz | 636000 | 3500.94 MHz | 30 kHz | 60,000 kHz | 166 RBs +8 REs |
| B | 3560–3600 MHz | 638668 | 3560.94 MHz | 30 kHz | 0 (cell B being used as a reference) | 0 RB+0 RE |

NOTE

- For details about Point A, see 3GPP TS 38.211 V16.5.0.
- The RIM-RS frequency-domain bandwidth is 20 MHz (48 RBs). Therefore, the cells between which mutual RIM-RS transmission and detection are performed must have at least 20 MHz overlapping bandwidths, and their SCSs must be the same.

## Enough/Not Enough Indication for Type-1 RIM-RS

The function of enough/not enough indication for type-1 RIM-RS is controlled by the **RIM_RS1_ENOUGH_IND_SW** option of the **NRDUCellRimConfig**.***RimRsFunctionSwitch*** parameter. After this function is enabled, the type-1 RIM-RS sent by the gNodeB serving the victim cell carries information about whether interference avoidance is performed to a sufficient extent.

When remote interference detection is performed in a victim cell (see step 1 in **Figure 4-1**):

- If the interference strength difference between uplink symbols is greater than the **NRDUCellRimConfig**.***UlPwrDiffThld*** parameter value, interference avoidance does not reach the expectation. In this case, the RIM-RS carries the "not enough" indication, instructing the interfering cell to continue or enhance remote interference avoidance.
- Otherwise, interference avoidance has reached the expectation. In this case, the RIM-RS carries the "enough" indication, instructing the interfering cell to stop remote interference avoidance or reduce the interference avoidance effect.

## Type-2 RIM-RS

The gNodeB generates an RIM-RS based on the following protocol-defined parameters. For details, see 3GPP TS 38.211 V16.5.0 and 3GPP TS 28.541 V16.1.0.

**Table 4-6** Type-2 RIM-RS configuration parameters

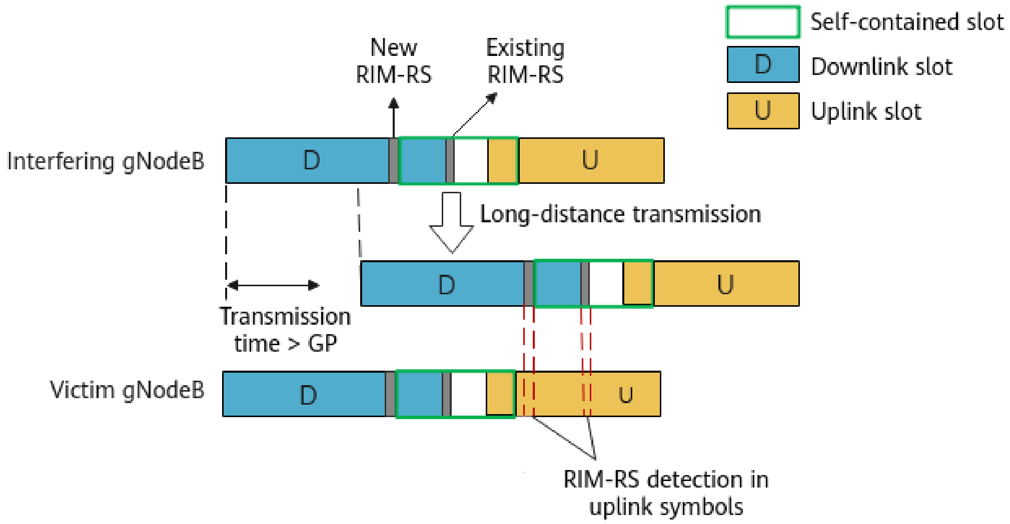
<table>
<tr><th>Parameter Name</th><th>Parameter ID</th><th>Function</th></tr>
<tr><td>RIM-RS Type2 Set ID Size</td><td>NRDUCellRimConfig.RimRsType2SetIdSize</td><td>This parameter determines the maximum number of interference sources that can be detected.</td></tr>
<tr><td>RIM-RS Type2 Set ID</td><td>NRDUCellRimConfig.RimRsType2SetId</td><td>This parameter uniquely identifies an interference source.</td></tr>
<tr><td>RIM-RS Type2 Sequence Candidate Number</td><td>NRDUCellRimConfig.RimRsType2SeqCandNum</td><td>This parameter indicates the number of type-2 RIM-RSs that can be transmitted on a frequency domain resource.</td></tr>
<tr><td>RIM-RS Type2 Scrambling ID for IS0 to RIM-RS Type2 Scrambling ID for IS7</td><td>NRDUCellRimConfig.RimRsType2ScidForIs0 to NRDUCellRimConfig.RimRsType2ScidForIs7</td><td>This parameter uniquely identifies a type-2 RIM-RS. If n RIM-RS candidate resources are configured, set the first n scrambling IDs to different effective values (0 to 1023).</td></tr>
<tr><td>RIM-RS Scramble Timer Multiplier Factor</td><td>NRDUCellRimConfig.RimRsScrTimerMultiFactor</td><td rowspan="2">The parameters are used to specify the scrambling timer factor. They together with the type-2 RIM-RS scrambling ID determine a RIM-RS. The two parameters are also used for type-1 RIM-RS.</td></tr>
<tr><td>RIM-RS Scramble Timer Offset</td><td>NRDUCellRimConfig.RimRsScrTimerOffset</td></tr>
</table>

**NRDUCellRimConfig**.***RimRsStartFreqOffset*** and **NRDUCellRimConfig**.***RimRsStartFreqReOffset*** jointly determine the start frequency-domain position for both type-2 and type-1 RIM-RSs. For details about how to configure the two parameters, see **Type-1 RIM-RS**.

## Near-Far Functionality for Type-1/Type-2 RIM-RS

Near-far functionality for type-1 and type-2 RIM-RS increases RIM-RS transmissions occasions in downlink slots, thereby enabling the gNodeB to detect interference sources at a longer distance (around 400 km as opposed to the original 200 km). The **RIM_RS1_NEAR_FAR_SW** and **RIM_RS2_NEAR_FAR_SW** options of the **NRDUCellRimConfig**.***RimRsFunctionSwitch*** parameter determine whether to enable near-far functionality for type-1 and near-far functionality for type-2 RIM-RS, respectively. As shown in **Figure 4-5**, the new and original RIM-RSs are of the same type.

**Figure 4-5** Near-far functionality for type-1/type-2 RIM-RS




## 4.1.2 Remote Interference Adaptive Avoidance

**Remote interference adaptive avoidance dynamically adjusts the GP based on the result of RIM-RS detection, thereby reducing the impact of remote interference on gNodeBs.** This function is controlled by the **RIM_ADAPT_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter. For details about the procedure of triggering and exiting remote interference adaptive avoidance, see **Figure 4-1**.

After detecting RIM-RS, the gNodeB serving the interfering cell measures the strength of remote interference experienced by the cell and identifies the corresponding symbol positions of the interference source. It then adjusts the GP of the cell based on these results. **Figure 4-7** shows an example of long-distance transmission with GP adjustment. This example assumes that the long-distance transmission has a delay of 15 symbols (see **Figure 4-6**). For details about the relationship between the interfering cell and the victim cell, see **Figure 4-1**.

**Figure 4-6** Implementation without GP adjustment

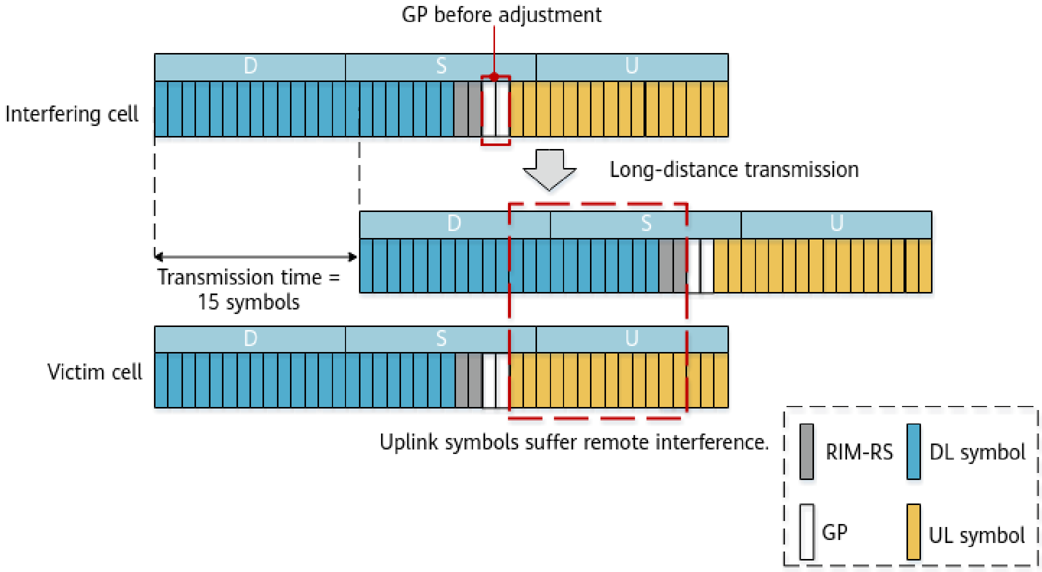
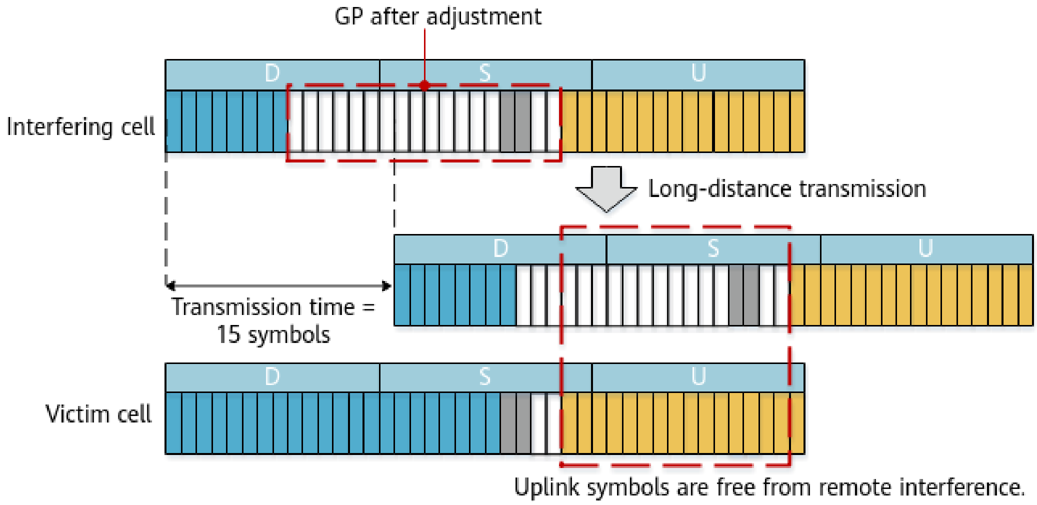



**Figure 4-7** Implementation with GP adjustment




Remote interference adaptive avoidance offers five GP adjustment levels, that is, five alternatives on the number of symbols added for the GP. The distance within which no interference is caused by the source of interference (interference-free distance for short) increases by about 10.7 km with the addition of each symbol. The GP adjustment level is specified by the **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter. This parameter can be set to an appropriate value depending on distances from interference sources. **Table 4-7** summarizes GP adjustment levels and the corresponding interference-free distance in case of single-period 8:2 slot configuration.

**Table 4-7** GP adjustment levels

| Parameter Value | GP Adjustment Level | Interference-free Distance (with Single-Period 8:2 Slot Configuration) |
|---|---|---|
| **SYMBOL_EXTEND** | Two additional symbols for the GP | Within 60 km |
| **FOUR_SYMBOL_EXTEND** | Four additional symbols for the GP | Within 80 km |
| **SLOT_S** | The GP is extended to include all non-uplink symbols of a self-contained slot. | Within 100 km |
| **DEFAULT** | 17 or 18 additional symbols for the GP | Within 200 km |
| **SLOT_D_D_S** | The GP is extended to include all non-uplink symbols of a self-contained slot and the two downlink slots that are immediately followed by the self-contained slot. | Within 400 km |

In GP adjustment, a larger number of additional symbols results in more downlink resource loss. Therefore, you are advised to consider the resource loss and distances from interference sources when setting the **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter.

NOTE

Different slot structures are required for different GP adjustment levels, as described below:

- **FOUR_SYMBOL_EXTEND**: The SS2, SS52, or SS82 slot structure is required.
- **SLOT_S** and **SYMBOL_EXTEND**: The SS2, SS4, SS54, or SS84 slot structure is required.
- **SLOT_D_D_S**: The SS54 slot structure is required.

If the value of the **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter is changed for a cell, the cell will exit the remote interference adaptive avoidance state. Moreover, the cell will re-evaluate the specific conditions to determine whether it should re-enter the state.

After the GP is adjusted, the number of synchronization signal and PBCH block (SSB) beams may decrease, affecting cell coverage. To ensure cell coverage, the broadcast channel power compensation function is introduced. This function is controlled by the **PWR_COMPENSATE_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter. After this option is selected and remote interference adaptive avoidance takes effect, the gNodeB compensates for the power of the broadcast channel by increasing the SSB power, thereby improving cell coverage.

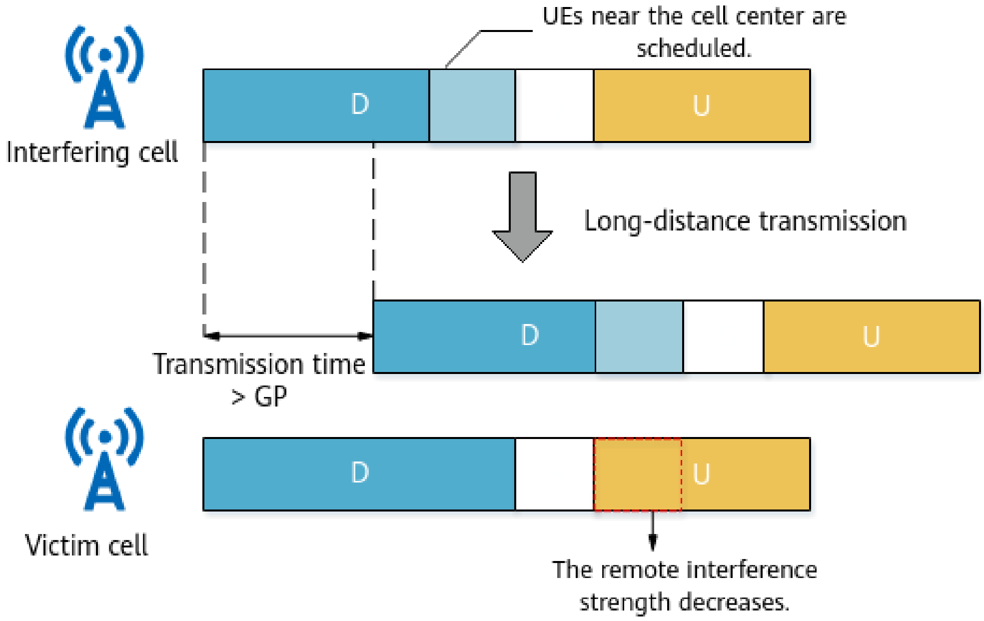
In scenarios where the GP cannot be dynamically adjusted due to RIM-RS detection unavailability, such as when inter-RAT intra-frequency remote interference occurs, the forcible GP adjustment function can be enabled to avoid interference to the remote end. This function is controlled by the **FORCED_RIM_AVOID_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter. After this function is enabled, the base station adjusts the GP of the cell by controlling the number of symbols used for scheduling, regardless of whether RIM-RS is detected in the cell. In addition, the cell is forced to remain in the adjusted state to reduce remote interference to other cells. This function does not change the cell's slot configuration (indicated by **NRDUCELL**.***SlotStructure***).

## 4.1.3 Enhanced Remote Interference Adaptive Avoidance

**Enhanced remote interference adaptive avoidance mitigates remote interference to victim cells by adjusting the time-domain positions for scheduling of different UEs in interfering cells.** This function is controlled by the **RIM_ADAPT_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter. For details about the procedure of triggering and exiting **enhanced remote interference adaptive avoidance**, see Figure 4-1.

- UEs at the center of interfering cells (referred to as UEs with low interference) impose little interference on victim cells, and those at the cell edge (referred to as UEs with high interference) impose great interference on victim cells. After enhanced remote interference adaptive avoidance takes effect, UEs with low interference are scheduled in downlink symbols close to the GP, and those with high interference are scheduled in other downlink symbols, as shown in Figure 4-8, thereby reducing remote interference to victim cells.
- If both remote interference adaptive avoidance and enhanced remote interference adaptive avoidance are enabled, the base station dynamically adjusts the GP and then adjusts the time domain scheduling ranges for UEs in the new slot configuration.

**Figure 4-8** Enhanced remote interference adaptive avoidance



# 4.2 Network Analysis

## 4.2.1 Benefits

### Deployment Suggestions

It is recommended that this function be enabled when mutual RIM-RS detection is available (that is, RIM-RS configurations are consistent) between the gNodeB serving the interfering cells and the gNodeB serving the victim cells.

It is recommended that remote interference adaptive avoidance and enhanced remote interference adaptive avoidance be enabled when accessibility KPIs deteriorate for a long time and the following conditions are met:

- **N.GAP.LastSymbol.Pwr** > **N.S.Symbol11.NI.Avg** > **N.S.Symbol12.NI.Avg** > **N.S.Symbol13.NI.Avg** > –107 dBm
- **N.GAP.LastSymbol.Pwr – N.UL.Last.Symbol13.Pwr** > 15 dB

Precautions for enabling the type-2 RIM-RS function are as follows:

- The status of the type-2 RIM-RS function switch must be the same for both the interfering and victim cells (for example, the switch is off for these cells). Otherwise, RIM-RS detection will fail.
- In 5G RAN V100R005C10 and earlier versions, the gNodeB does not support type-2 RIM-RS. If type-2 RIM-RS is required, the gNodeBs serving the interfering and victim cells must be upgraded.
- The type-2 RIM-RS function does not affect the RIM-RS content. When only type-1 RIM-RS is used, it takes shorter time for gNodeBs to complete RIM-RS detection, and the performance is better. Therefore, it is recommended that the type-2 RIM-RS function be disabled except in special scenarios (for example, when the RIM-RS configurations of other vendors' devices are different from those of Huawei devices).

Precautions for enabling the function of enough/not enough indication for type-1 RIM-RS are as follows:

- The status of the switch for enough/not enough indication for type-1 RIM-RS must be the same for both the interfering and victim cells (for example, the switch is off for these cells). Otherwise, RIM-RS detection will fail.
- In versions earlier than 5G RAN V100R006C10, gNodeBs do not support the function of enough/not enough indication for type-1 RIM-RS. If this function is required, the gNodeBs serving the interfering and victim cells must be upgraded.
- When the function of enough/not enough indication for type-1 RIM-RS is disabled for cells, gNodeBs complete RIM-RS detection in a shorter time, thereby delivering better performance. Therefore, you are advised not to enable this function except in special scenarios (for example, scenarios where this function is enabled for other vendors' devices).

NOTE

- Whether remote interference management should be enabled depends on the actual interference on the network. Considering the reciprocity of the atmospheric ducts, you are advised to enable remote interference management for all gNodeBs that reside in areas subjected to atmospheric duct interference.
- You are advised to enable the function in **5 Adaptive Interference Suppression** when mutual RIM-RS detection is not supported between the gNodeB serving the interfering cells and the gNodeB serving the victim cells.

### Benefits

After remote interference adaptive avoidance and enhanced remote interference adaptive avoidance are enabled, the gNodeB automatically adjusts the GP and schedules the UEs with high interference in certain downlink slots far away from the uplink symbols. This mitigates the impact of remote interference on the gNodeB, thereby improving network performance and restoring basic services of cells that suffer interference. For details about indicators involved, see **4.4.3 Network Monitoring**.

## 4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

- Remote interference adaptive avoidance
  - During GP adjustment of remote interference adaptive avoidance (including GP expansion and restoration):
    - During dynamic adjustment of the number of SSB beams, no dedicated preamble is allocated. In this case, SA/NSA/SA SCC UEs switch from the non-contention-based random access (RA) mode to the contention-based RA mode in scenarios such as incoming handovers and downlink resynchronization, and the incoming handover success rate may decrease.
    - During dynamic adjustment of the number of SSB beams, the cell coverage deteriorates or fluctuates within a short period of time, the downlink receive level of UEs deteriorates, and the average downlink UE throughput decreases. In the case of GP restoration, the preceding impacts disappear after beam adjustment is complete.
    - During dynamic adjustment of the number of SSB beams, beam information needs to be reconfigured for UEs. Before the beam reconfiguration is complete, the update of the beams for UEs to receive paging messages is delayed. As a result, some UEs may fail to receive paging messages.
    - System information is reconfigured for NSA UEs, causing the UEs to re-initiate RA.
    - Dynamic online adjustment of the GP and SSB beam quantity requires UE support. UEs may experience service drops if they are incapable of dynamic online adjustment of the GP and SSB beam quantity.

- After remote interference adaptive avoidance takes effect:
  - After the GP is adjusted, the number of symbols available for downlink scheduling or the average number of available downlink physical resource blocks (PRBs) decreases, the total number of downlink PDSCH transmission time intervals (TTIs) decreases, the downlink peak rate and average downlink throughput of the cell decrease, and the delay increases slightly.
  - After the GP is adjusted, the number of SSB beams decreases. If broadcast channel power compensation is enabled, the SSB coverage remains unchanged; otherwise, the SSB edge coverage will be adversely affected. When more than two NR carriers are supported by an RF module, the power aggregation capability of the base station decreases, the SSB power that can be compensated for decreases, and the SSB edge coverage may be adversely affected.
  - After the GP is adjusted, the number of SSB beams corresponding to the Remaining Minimum SI (RMSI) decreases. As a result, the RMSI cannot be sent in slots close to the extended GP, and the time-frequency domain resources occupied by the RMSI decrease. The average number of used downlink PRBs and the average number of PRBs used by signaling on the PDSCH decrease.
  - After the number of SSB beams decreases, the number of SSB beams in the local cell may be inconsistent with that in a neighboring cell. As a result, the common channel of the neighboring cell causes intra-frequency interference to the PDSCH of the local cell, severely affecting the average downlink cell throughput. In this case, the common channel interference avoidance function can be enabled to reduce the impact of interference. This function is controlled by the **SSB_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter. For details, see section "Common Channel Interference Avoidance" in *Interference Avoidance*.
  - After broadcast channel power compensation is enabled, the broadcast channel power may converge, the resources used for downlink scheduling may decrease, and the downlink peak throughput of the cell and UEs may decrease.
  - For UEs that perform full-buffer services, UE-level CSI-RS beams take effect before remote interference adaptive avoidance takes effect, and cell-level CSI-RS beams take effect after remote interference adaptive avoidance takes effect. In this case, the downlink throughput decreases significantly.
  - When the **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter is set to **SYMBOL_EXTEND** or **FOUR_SYMBOL_EXTEND**, the number of symbols in a self-contained slot decreases. If the DRX On Duration period is fixed at a specific value, the self-contained slot may not accommodate the TB size for scheduling in a BWP2 or RedCap scenario. As such, the self-contained slot does not have an opportunity for being scheduled and the delay increases. If this situation persists for a long time, service drops may occur. It is

recommended that the fixed DRX On Duration period be increased to be longer than 8 ms in gap expansion scenarios.

- When 8:2 slot configuration is used and the **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter is set to **SYMBOL_EXTEND** or **FOUR_SYMBOL_EXTEND**, the self-contained slot structure will be 4:6:4 after gap expansion. In addition, if two symbols are selected in PDCCH symbol adaptation, the start and length indicator value (SLIV) determined for scheduling in the self-contained slot will be changed to specific values (S = 2, L = 2), in which case type A scheduling is not supported but only type B scheduling is supported. If a UE does not support type B scheduling, scheduling in the self-contained slot fails and the downlink throughput of the UE decreases.
- After the GP is adjusted, the number of downlink symbols in a self-contained slot is less than 7. A few UEs may not support scheduling in which the number of symbols is less than 7. As a result, scheduling in the self-contained slot fails.

- Enhanced remote interference adaptive avoidance

  After enhanced remote interference adaptive avoidance takes effect, UEs with high interference are less likely to be scheduled, the downlink peak rate and average throughput of massive MIMO cells do not change greatly, and the downlink peak rate and average throughput of non-massive-MIMO cells decrease.
- Type-2 RIM-RS

  The network waits longer before it starts remote interference adaptive avoidance or enhanced remote interference adaptive avoidance after RIM-RS detection.
- Enough/Not enough indication for type-1 RIM-RS

  The RIM-RS transmission period is prolonged. As a result, the network waits longer before it starts remote interference adaptive avoidance or enhanced remote interference adaptive avoidance.

The indicators involved in the preceding impacts are as follows:

| Indicator Description | Indicator Name |
|---|---|
| Average downlink UE throughput | **User Downlink Average Throughput (DU)** |
| Average downlink cell throughput | **Cell Downlink Average Throughput (DU)** |
| Average number of available downlink PRBs | **N.PRB.DL.Avail.Avg** |
| Total number of downlink PDSCH TTIs | **N.DL.PDSCH.Tti.Num** |
| Average number of used downlink PRBs | **N.PRB.DL.Used.Avg** |

| Indicator Description | Indicator Name |
|---|---|
| Average number of PRBs used by signaling on the PDSCH | **N.PRB.DL.SrbUsed.Avg** |

# 4.3 Requirements

## 4.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-030208 | Remote Interference Management (RIM) | NR0S00RIMG00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

## 4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 4.3.2.1 Prerequisite Functions

#### Remote Interference Adaptive Avoidance

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Synchronization | None | *Synchronization* | Remote interference management requires that the GNSS standard second value be obtained through the synchronization or network synchronization function. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Network synchronization | None | *Synchronization* | Remote interference management requires that the GNSS standard second value be obtained through the synchronization or network synchronization function. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| 3D coverage pattern (low-frequency TDD) | **NRDUCellTrpBeam**.***CoverageScenario*** | *Beam Management* | When remote interference adaptive avoidance is enabled, the ***CoverageScenario*** parameter can only be set to **DEFAULT**, a value in the range of **SCENARIO_1** to **SCENARIO_16**, **SCENARIO_36**, or **CUSTOMIZED_BEAM_SCENARIO**.<br>Remote interference adaptive avoidance takes effect only when the **Coverage Scenario** parameter is set to **CUSTOMIZED_BEAM_SCENARIO** and the following conditions are met. Otherwise, only RIM-RS transmission and detection can be performed.<br>● In scenarios where the slot configuration is single-period 8:2:<br>– The **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter is set to **SLOT_D_D_S**, and up to five beams are configured by using the **ADD NRDUCELLTRPCUSTBEAM** command.<br>– The **NRDUCellRimConfig**.***IntrfAvoid*** |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| | | | ***DlExtensionLevel*** parameter is set to **DEFAULT**, and up to six beams are configured by using the **ADD NRDUCELLTRPCUSTBEAM** command.<br>– The **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter is set to any other value, and up to seven beams are configured by using the **ADD NRDUCELLTRPCUSTBEAM** command.<br>● In scenarios where the slot configuration is 4:1:<br>– The **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter is set to **SYMBOL_EXTEND**, and up to seven beams are configured by using the **ADD NRDUCELLTRPCUSTBEAM** command.<br>– The **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter is set to **DEFAULT**, and up to three |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| | | | beams are configured by using the **ADD NRDUCELLTRPCUSTBEAM** command.<br>– The **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter is set to any other value, and up to five beams are configured by using the **ADD NRDUCELLTRPCUSTBEAM** command.<br>● In scenarios where the slot configuration is dual-period 8:2:<br>– The **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter is set to **SYMBOL_EXTEND**, and up to four beams are configured by using the **ADD NRDUCELLTRPCUSTBEAM** command.<br>– The **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter is set to **DEFAULT**, and up to three beams are configured by using the **ADD** |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| | | | **NRDUCELLTRPCUSTBEAM** command.<br>– The **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter is set to any other value, and up to five beams are configured by using the **ADD NRDUCELLTRPCUSTBEAM** command. |

## Enhanced Remote Interference Adaptive Avoidance

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Synchronization | None | *Synchronization* | Remote interference management requires that the GNSS standard second value be obtained through the synchronization or network synchronization function. |
| Network synchronization | None | *Synchronization* | Remote interference management requires that the GNSS standard second value be obtained through the synchronization or network synchronization function. |

### Type-2 RIM-RS

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Remote interference adaptive avoidance | **RIM_ADAPT_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter | *Remote Interference Management (Low-Frequency TDD)* | The type-2 RIM-RS function can be enabled only when remote interference adaptive avoidance or enhanced remote interference adaptive avoidance is enabled. |
| Enhanced remote interference adaptive avoidance | **RIM_ADAPT_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter | *Remote Interference Management (Low-Frequency TDD)* | |

### Enough/Not Enough Indication for Type-1 RIM-RS

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Remote interference adaptive avoidance | **RIM_ADAPT_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter | *Remote Interference Management (Low-Frequency TDD)* | The enough/not enough indication for type-1 RIM-RS function can be enabled only when remote interference adaptive avoidance or enhanced remote interference adaptive avoidance is enabled and the **NRDUCellRimConfig**.***RimRsType1SeqCandNum*** parameter is set to a value other than **SEQ_NUM_1**. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Enhanced remote interference adaptive avoidance | **RIM_ADAPT_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter | *Remote Interference Management (Low-Frequency TDD)* | The enough/not enough indication for type-1 RIM-RS function can be enabled only when remote interference adaptive avoidance or enhanced remote interference adaptive avoidance is enabled and the **NRDUCellRimConfig**.***RimRsType1SeqCandNum*** parameter is set to a value other than **SEQ_NUM_1**. |

### 4.3.2.2 Mutually Exclusive Functions

#### Remote Interference Adaptive Avoidance

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Data Transmission Mode | **NRDUCellMultiTrp**.***DataTransMode*** | *Cell Combination* | When remote interference adaptive avoidance is enabled, the **NRDUCellMultiTrp**.***DataTransMode*** parameter cannot be set to **JOINT_MODE**. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| eDRX in idle mode | **IDLE_MODE_EDRX_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter | *RedCap* | The two functions are mutually exclusive: eDRX in idle mode and remote interference adaptive avoidance. |
| Paging Capacity Mode | **NRDUCellPagingConfig**.***PagingCapacityMode*** | *5G Networking and Signaling* | If remote interference adaptive avoidance is enabled, the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter can only be set to **DEFAULT_MODE**. |
| Time-Domain Reservation Mode | **NRDUCellTimeIntrf**.***TimeDomainRsvMode*** | None | If remote interference adaptive avoidance is enabled and the **NRDUCellTimeIntrf**.***TimeDomainRsvMode*** parameter is set to **TIME_DOMAIN_RESERVED_MODE_1**, **TIME_DOMAIN_RESERVED_MODE_2**, or **TIME_DOMAIN_RESERVED_MODE_3**, the **NRDUCellRimConfig**.***IntrfAvoidDlExtensionLevel*** parameter cannot be set to **DEFAULT**. |
| PLLC Primary Cell Switch | **PLLC_PRIMARY_CELL_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter | None | The two switches cannot be both enabled: **PLLC Primary Cell Switch** and the switch for remote interference adaptive avoidance. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | All the following conditions must be met if remote interference adaptive avoidance and Hyper Cell need to be both enabled:<br>● The **FORCED_RIM_AVOID_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter is not selected.<br>● The **NRDUCellRimConfig**.***IntrfAvoidDetectPrdNum*** parameter is not set to **0**.<br>● The **NRDUCellRimConfig**.***RimRsType1IntrfThld*** parameter is set to **-50**. |
| Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** | *Cell Combination* | All the following conditions must be met if remote interference adaptive avoidance and Cell Combination need to be both enabled:<br>● The **FORCED_RIM_AVOID_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter is not selected.<br>● The **NRDUCellRimConfig**.***IntrfAvoidDetectPrdNum*** parameter is not set to **0**.<br>● The **NRDUCellRimConfig**.***RimRsType1IntrfThld*** parameter is set to **-50**. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED** | *High Speed Mobility* | All the following conditions must be met if remote interference adaptive avoidance and High-speed Railway Superior Experience need to be both enabled:<br>● The **FORCED_RIM_AVOID_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter is not selected.<br>● The **NRDUCellRimConfig**.***IntrfAvoidDetectPrdNum*** parameter is not set to **0**.<br>● The **NRDUCellRimConfig**.***RimRsType1IntrfThld*** parameter is set to **-50**. |
| VMIMO in a combined-RRU 4T4R cell (low-frequency TDD) | **NRDUCellAlgoSwitch**.***VmimoSwitch*** | *Indoor Coverage* | The two functions are mutually exclusive: VMIMO in a combined-RRU 4T4R cell and remote interference adaptive avoidance. |
| Extended Cell Range (TDD) | **NRDUCell**.***CellRadius*** set to a value greater than **14500** | *Extended Cell Range* | The two functions are mutually exclusive: remote interference adaptive avoidance and Extended Cell Range. |
| Extended Cell Range of 90 km (TDD) | **NRDUCell**.***CellRadius*** set to a value greater than **60000** and less than or equal to **90000** | *Extended Cell Range* | The two functions are mutually exclusive: remote interference adaptive avoidance and Extended Cell Range of 90 km. |
| Interfering cell detection | **NRDUCellTimeIntrf**.***CrossLinkIntrfCfg*** set to **CROSS_LINK_SEQ_S_D_TYPE1** | *Standards Compliance* | The two functions are mutually exclusive: interfering cell detection and remote interference adaptive avoidance. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| IAB deployment | **NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB** | None | When the **NRDUCell**.***DeployPosition*** parameter is set to **DEPLOY_IAB**, remote interference adaptive avoidance must be disabled. |
| Intra-Frequency Assistant Cell | **NRDUCell**.***SupplementaryCellInd*** set to **INTRA_FREQ_ASSISTANT_CELL** | None | When the **NRDUCell**.***SupplementaryCellInd*** parameter is set to **INTRA_FREQ_ASSISTANT_CELL**, all these options must be deselected: (1) **RIM_ADAPT_AVOID_SW**, **RIM_ADAPT_AVOID_ENH_SW**, and **RIM_ADAPT_SUPPRESSION_SW** of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter, and (2) **INTRA_GNB_SU_COMM_SW** and **INTER_CELL_INTRF_AVOID_SW** of the **NRDuCellCollabserv**.***MultiCellMimoSwitch*** parameter. |
| Low Latency High Reliability Level 1 Sw | **LOW_LAT_HIGH_RELIAB_LEVEL1_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter | None | When the **LOW_LAT_HIGH_RELIAB_LEVEL1_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter is selected, remote interference adaptive avoidance must be disabled. |
| Distributed massive MIMO | **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter | *Distributed Antenna Solutions (Low-Frequency TDD)* | The two functions are mutually exclusive: remote interference adaptive avoidance and distributed massive MIMO. |

## Enhanced Remote Interference Adaptive Avoidance

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Data Transmission Mode | **NRDUCellMultiTrp**.***DataTransMode*** | *Cell Combination* | When enhanced remote interference adaptive avoidance is enabled, the **NRDUCellMultiTrp**.***DataTransMode*** parameter cannot be set to **JOINT_MODE**. |
| Paging Capacity Mode | **NRDUCellPagingConfig**.***PagingCapacityMode*** | *5G Networking and Signaling* | If enhanced remote interference adaptive avoidance is enabled, the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter can only be set to **DEFAULT_MODE**. |
| PLLC Primary Cell Switch | **PLLC_PRIMARY_CELL_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter | None | The two switches cannot be both enabled: **PLLC Primary Cell Switch** and the switch for enhanced remote interference adaptive avoidance. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | All the following conditions must be met if enhanced remote interference adaptive avoidance and Hyper Cell need to be both enabled:<br>● The **FORCED_RIM_AVOID_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter is not selected.<br>● The **NRDUCellRimConfig**.***IntrfAvoidDetectPrdNum*** parameter is not set to **0**.<br>● The **NRDUCellRimConfig**.***RimRsType1IntrfThld*** parameter is set to **-50**. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** | *Cell Combination* | All the following conditions must be met if enhanced remote interference adaptive avoidance and Cell Combination need to be both enabled:<br>● The **FORCED_RIM_AVOID_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter is not selected.<br>● The **NRDUCellRimConfig**.***IntrfAvoidDetectPrdNum*** parameter is not set to **0**.<br>● The **NRDUCellRimConfig**.***RimRsType1IntrfThld*** parameter is set to **-50**. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED** | *High Speed Mobility* | All the following conditions must be met if enhanced remote interference adaptive avoidance and High-speed Railway Superior Experience need to be both enabled:<br>● The **FORCED_RIM_AVOID_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter is not selected.<br>● The **NRDUCellRimConfig**.***IntrfAvoidDetectPrdNum*** parameter is not set to **0**.<br>● The **NRDUCellRimConfig**.***RimRsType1IntrfThld*** parameter is set to **-50**. |
| VMIMO in a combined-RRU 4T4R cell (low-frequency TDD) | **NRDUCellAlgoSwitch**.***VmimoSwitch*** | *Indoor Coverage* | The two functions are mutually exclusive: VMIMO in a combined-RRU 4T4R cell and enhanced remote interference adaptive avoidance. |
| Extended Cell Range (TDD) | **NRDUCell**.***CellRadius*** set to a value greater than **14500** | *Extended Cell Range* | The two functions are mutually exclusive: enhanced remote interference adaptive avoidance and Extended Cell Range. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Extended Cell Range of 90 km (TDD) | **NRDUCell**.***CellRadius*** set to a value greater than **60000** and less than or equal to **90000** | *Extended Cell Range* | The two functions are mutually exclusive: enhanced remote interference adaptive avoidance and Extended Cell Range of 90 km. |
| Interfering cell detection | **NRDUCellTimeIntrf**.***CrossLinkIntrfCfg*** set to **CROSS_LINK_SEQ_S_D_TYPE1** | *Standards Compliance* | The two functions are mutually exclusive: interfering cell detection and enhanced remote interference adaptive avoidance. |
| IAB deployment | **NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB** | None | When the **NRDUCell**.***DeployPosition*** parameter is set to **DEPLOY_IAB**, enhanced remote interference adaptive avoidance must be disabled. |
| Intra-Frequency Assistant Cell | **NRDUCell**.***SupplementaryCellInd*** set to **INTRA_FREQ_ASSISTANT_CELL** | None | When the **NRDUCell**.***SupplementaryCellInd*** parameter is set to **INTRA_FREQ_ASSISTANT_CELL**, all these options must be deselected: (1) **RIM_ADAPT_AVOID_SW**, **RIM_ADAPT_AVOID_ENH_SW**, and **RIM_ADAPT_SUPPRESSION_SW** of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter, and (2) **INTRA_GNB_SU_COMM_SW** and **INTER_CELL_INTRF_AVOID_SW** of the **NRDuCellCollabserv**.***MultiCellMimoSwitch*** parameter. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Low Latency High Reliability Level 1 Sw | **LOW_LAT_HIGH_RELIAB_LEVEL1_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter | None | When the **LOW_LAT_HIGH_RELIAB_LEVEL1_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter is selected, enhanced remote interference adaptive avoidance must be disabled. |
| Distributed massive MIMO | **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter | *Distributed Antenna Solutions (Low-Frequency TDD)* | The two functions are mutually exclusive: enhanced remote interference adaptive avoidance and distributed massive MIMO. |

### 4.3.2.3 Function Impacts

## Remote Interference Adaptive Avoidance

- Functions related to energy saving:

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Uplink symbol power saving | **UL_SYMBOL_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Energy Conservation and Emission Reduction* | When remote interference adaptive avoidance is in effect, the benefits of uplink symbol power saving decrease due to the reduced power saving duration. In this scenario, the **RIM_UL_SYMBOL_SHUTDOWN_OPT_SW** option of the **NRDUCellPwrSaveAlgo**.***PowerSavingAlgoSwitch*** parameter can be selected so that uplink symbol power saving can take effect in a higher proportion of cases, increasing energy saving gains. |
| LNR coordinated channel shutdown | **MULTI_RAT_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Multi-RAT Coordinated Energy Saving* | LNR coordinated channel shutdown does not take effect when remote interference adaptive avoidance is in effect. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| NR low power consumption mode | **LOW_PWR_MODE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Green Site* | When both low power consumption mode and remote interference adaptive avoidance are enabled and a customized backup power saving policy is used in low power consumption mode, the following rules apply:<br>● If the coverage compensation policy in NR low power consumption mode is disabled (by deselecting the **COV_COMPENSATION_POLICY_SW** option of the **NRDUCellPowerSaving**.***LowPwrModeAlgoSwitch*** parameter), channel shutdown in low power consumption mode does not take effect.<br>● If the coverage compensation policy in NR low power consumption mode is enabled, low power consumption mode and remote interference adaptive avoidance can both take effect. In this case, coverage performance of low power consumption mode may deteriorate. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| RF channel intelligent shutdown | **RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Energy Conservation and Emission Reduction* | When remote interference adaptive avoidance is in effect, RF channel intelligent shutdown cannot take effect. After atmospheric ducts disappear, RF channel intelligent shutdown can take effect. |
| Energy saving based on flexible frequency-domain scheduling | **FLEX_FREQ_SCH_ENERGY_SAVING_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Energy Conservation and Emission Reduction* | When both remote interference adaptive avoidance and energy saving based on flexible frequency-domain scheduling take effect, the gains of the latter decrease. |
| TTI-level channel shutdown | **TTI_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Energy Conservation and Emission Reduction* | When both remote interference adaptive avoidance and TTI-level channel shutdown take effect, the energy saving gains of the latter decrease. |

- Functions related to interference suppression:

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Proactive avoidance of remote interference | **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter | *Remote Interference Management (Low-Frequency TDD)* | When the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter are selected, the measured value of **N.UL.Last.Symbol13.Pwr** may be greater than expected, affecting the interference detection results of remote interference management and clock out-of-synchronization detection. |
| SRS Intrf Avoid PUSCH Sch Sw | **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter | None | When the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected, the measured value of **N.GAP.LastSymbol.Pwr** may be greater than expected, affecting the interference detection results of remote interference management and clock out-of-synchronization detection. |

- Functions related to user-centric network (UCN):

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Intra-base-station DL CoMP | **INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter | *CoMP* | When both intra-base-station DL CoMP and remote interference adaptive avoidance are enabled, the resulting impacts are as follows: Intra-base-station DL CoMP cannot take effect in self-contained slots if two specific conditions are met—(1) in both the serving and cooperating cells, the slots include downlink symbols and (2) downlink symbols in the slots of the serving cell outnumber those of the cooperating cell. However, intra-base-station DL CoMP can take effect in self-contained slots if the two conditions are not met. |
| Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | After remote interference adaptive avoidance is enabled, hyper cells perform only RIM-RS transmission and detection to trigger interference avoidance in common cells. They do not adjust the GP due to heavy traffic, which helps reduce resource consumption. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** | *Cell Combination* | After remote interference adaptive avoidance is enabled, combined cells perform only RIM-RS transmission and detection to trigger interference avoidance in common cells. They do not adjust the GP due to heavy traffic, which helps reduce resource consumption. |

- Other functions:

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Clock out-of-synchronization detection | **CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter | *Synchronization* | When a cell is sending silence sequences, it stops sending RIM-RSs. When a cell is sending out-of-synchronization sequences, it stops sending RIM-RSs. When a cell is detecting out-of-synchronization sequences, it stops detecting RIM-RSs. |
| Uplink intra-gNodeB intra-FR inter-band CA | **INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter | *Carrier Aggregation* | When remote interference adaptive avoidance is in effect, the multiple timing advances (MTA) function does not take effect for a UE in the uplink CA state. This means that the UE can use only a single TA for uplink timing. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Virtual Grid-based Multi-Frequency Coordination | • RSRP prediction model: **VG_MODEL_ALLOW_BUILD_FLAG** option of the **NRCellFreqRelation**.***AggregationAttribute*** parameter and **NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter<br>• Downlink intra-frequency spectral efficiency prediction model: **DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter and **DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter<br>• Intra-frequency SRS beam prediction model: **INTRAF_SRS_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter | *Virtual Grid-based Multi-Frequency Coordination* | • If the **NR_VGRID_AND_RIM_AVD_COLLAB_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is deselected and remote interference adaptive avoidance is in effect, ongoing building of virtual grid models will be suspended, and existing virtual grid models will go offline and become unavailable for use.<br>• If the **NR_VGRID_AND_RIM_AVD_COLLAB_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is selected for both the local and neighboring base stations and remote interference adaptive avoidance is in effect, virtual grid models can be used. However, they can only be used in the following two scenarios:<br>– The **FORCED_RIM_AVOID_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| | | | parameter is selected.<br>– The maximum number of SSB beams remains unchanged. |
| Combined-RRU cell | • **NRDUCellTrp**.***TxRxMode***<br>• **NRDUCellCoverage**.***SectorEqmId*** | *Cell Management* | In a combined-RRU cell, the two antennas involved may cover different directions. Some of the antennas may not receive RIM-RSs because remote interference comes from a specific direction. Therefore, a combined-RRU cell delivers worse RIM-RS detection performance compared to a common cell with the same number of antennas. |
| High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED** | *High Speed Mobility* | After remote interference adaptive avoidance is enabled, high-speed cells perform only RIM-RS transmission and detection to trigger interference avoidance in common cells. They do not adjust the GP due to heavy traffic, which helps reduce resource consumption. |

### Enhanced Remote Interference Adaptive Avoidance

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Uplink symbol power saving | **UL_SYMBOL_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter | *Energy Conservation and Emission Reduction* | When enhanced remote interference adaptive avoidance is in effect, the benefits of uplink symbol power saving decrease due to the reduced power saving duration. In this scenario, the **RIM_UL_SYMBOL_SHUTDOWN_OPT_SW** option of the **NRDUCellPwrSaveAlgo**.***PowerSavingAlgoSwitch*** parameter can be selected so that uplink symbol power saving can take effect in a higher proportion of cases, increasing energy saving gains. |
| Single DCI | **DL_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter | *Multi-Frequency Smart Aggregation* | Enhanced remote interference adaptive avoidance does not take effect for UEs for which single DCI has taken effect. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| NR low power consumption mode | **LOW_PWR_MODE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Green Site* | When both low power consumption mode and enhanced remote interference adaptive avoidance are enabled and a customized backup power saving policy is used in low power consumption mode, the following rules apply:<br>● If the coverage compensation policy in NR low power consumption mode is disabled (by deselecting the **COV_COMPENSATION_POLICY_SW** option of the **NRDUCellPowerSaving**.***LowPwrModeAlgoSwitch*** parameter), channel shutdown in low power consumption mode does not take effect.<br>● If the coverage compensation policy in NR low power consumption mode is enabled, low power consumption mode and enhanced remote interference adaptive avoidance can both take effect. In this case, coverage performance of low power consumption mode may deteriorate. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Clock out-of-synchronization detection | **CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter | *Synchronization* | When a cell is sending silence sequences, it stops sending RIM-RSs. When a cell is sending out-of-synchronization sequences, it stops sending RIM-RSs. When a cell is detecting out-of-synchronization sequences, it stops detecting RIM-RSs. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Virtual Grid-based Multi-Frequency Coordination | • RSRP prediction model: **VG_MODEL_ALLOW_BUILD_FLAG** option of the **NRCellFreqRelation**.***AggregationAttribute*** parameter and **NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter<br>• Downlink intra-frequency spectral efficiency prediction model: **DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter and **DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter<br>• Intra-frequency SRS beam prediction model: **INTRAF_SRS_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter | *Virtual Grid-based Multi-Frequency Coordination* | • If the **NR_VGRID_AND_RIM_AVD_COLLAB_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is deselected and enhanced remote interference adaptive avoidance is in effect, ongoing building of virtual grid models will be suspended, and existing virtual grid models will go offline and become unavailable for use.<br>• If the **NR_VGRID_AND_RIM_AVD_COLLAB_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is selected for both the local and neighboring base stations and enhanced remote interference adaptive avoidance is in effect, virtual grid models can be used. However, they can only be used in the following two scenarios:<br>– The **FORCED_RIM_AVOID_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| | | | parameter is selected.<br>– The maximum number of SSB beams remains unchanged. |
| Combined-RRU cell | • **NRDUCellTrp**.***TxRxMode***<br>• **NRDUCellCoverage**.***SectorEqmId*** | *Cell Management* | In a combined-RRU cell, the two antennas involved may cover different directions. Some of the antennas may not receive RIM-RSs because remote interference comes from a specific direction. Therefore, a combined-RRU cell delivers worse RIM-RS detection performance compared to a common cell with the same number of antennas. |
| Proactive avoidance of remote interference | **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter | *Remote Interference Management (Low-Frequency TDD)* | When the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter are selected, the measured value of **N.UL.Last.Symbol13.Pwr** may be greater than expected, affecting the interference detection results of remote interference management and clock out-of-synchronization detection. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| SRS Intrf Avoid PUSCH Sch Sw | **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter | None | When the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected, the measured value of **N.GAP.LastSymbol.Pwr** may be greater than expected, affecting the interference detection results of remote interference management and clock out-of-synchronization detection. |
| Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | After enhanced remote interference adaptive avoidance is enabled, hyper cells perform only RIM-RS transmission and detection to trigger interference avoidance in common cells. They do not adjust time-domain UE scheduling positions due to heavy traffic, which helps reduce resource consumption. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** | *Cell Combination* | After enhanced remote interference adaptive avoidance is enabled, combined cells perform only RIM-RS transmission and detection to trigger interference avoidance in common cells. They do not adjust time-domain UE scheduling positions due to heavy traffic, which helps reduce resource consumption. |
| High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED** | *High Speed Mobility* | After enhanced remote interference adaptive avoidance is enabled, high-speed cells perform only RIM-RS transmission and detection to trigger interference avoidance in common cells. They do not adjust time-domain UE scheduling positions due to heavy traffic, which helps reduce resource consumption. |

## 4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

All NR TDD-capable 2T2R, 4T4R, 8T8R, 32T32R, and 64T64R RF modules support this function.

## 4.3.4 Cells

The cell bandwidth is 20 MHz to 100 MHz, and the SCS is 30 kHz.

Remote interference adaptive avoidance supports the following slot structures (**NRDUCell**.***SlotStructure***) and slot configurations (**NRDUCell**.***SlotAssignment***):

- Slot structures **SS1** to **SS6** and **SS18**, with the 4:1 slot configuration
- Slot structures **SS51** to **SS56** and **SS518**, with the single-period 8:2 slot configuration
- Slot structures **SS81** to **SS86** and **SS818**, with the dual-period 8:2 slot configuration

Enhanced remote interference adaptive avoidance supports the following slot structures and slot configurations:

- Slot structures **SS1** to **SS6**, with the 4:1 slot configuration
- Slot structures **SS51** to **SS56**, with the single-period 8:2 slot configuration
- Slot structures **SS81** to **SS86**, with the dual-period 8:2 slot configuration

## 4.3.5 Others

- Remote interference adaptive avoidance requires that UEs support online dynamic adjustment of the GP as well as the number of SSB beams.
- When remote interference adaptive avoidance and enhanced remote interference adaptive avoidance are deployed in cells with different bandwidths (for example, in the scenario where cells are deployed across different operators), these cells must have an overlapping bandwidth of more than 20 MHz.

# 4.4 Operation and Maintenance

## 4.4.1 Data Configuration

### 4.4.1.1 Data Preparation

- Type-1 RIM-RS

  **Table 4-8** and **Table 4-9** describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment.

**Table 4-8** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM-RS Type1 Set ID Size | **NRDUCellRimConfig**.***RimRsType1SetIdSize*** | Set this parameter based on the number of cells for RIM-RS detection. The value of this parameter must be the same for all cells. |
| RIM-RS Type1 Set ID | **NRDUCellRimConfig**.***RimRsType1SetId*** | It is recommended that this parameter be set to different values for cells where this function is enabled. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM-RS Frequency Candidate Number | **NRDUCellRimConfig**.***RimRsFreqCandNum*** | Use the recommended value. Ensure that the parameter setting is the same among all of the cells.<br>If the interfering and victim cells differ in bandwidth (bandwidth size or frequency-domain position), set this parameter to its recommended value based on the overlapped bandwidth.<br>● For a DU cell, if the **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_20M** or **CELL_BW_30M**, the **NRDUCellRimConfig**.***RimRsFreqCandNum*** parameter must be set to **FREQ_NUM_1**.<br>● For a DU cell, if the **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_40M**, **CELL_BW_50M**, **CELL_BW_60M**, or **CELL_BW_70M**, the **NRDUCellRimConfig**.***RimRsFreqCandNum*** parameter must be set to **FREQ_NUM_1** or **FREQ_NUM_2**.<br>● For a DU cell, if the **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_80M**, **CELL_BW_90M**, or **CELL_BW_100M**, the **NRDUCellRimConfig**.***RimRsFreqCandNum*** parameter must be set to **FREQ_NUM_1**, **FREQ_NUM_2**, or **FREQ_NUM_4**. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM-RS Type1 Sequence Candidate Number | **NRDUCellRimConfig**.***RimRsType1SeqCandNum*** | Use the recommended value. Ensure that the parameter setting is the same among all of the cells.<br>● Scenario with slot assignment 8:2:<br>– For a DU cell, if the **NRDUCellRimConfig**.***RimRsFreqCandNum*** parameter is set to **FREQ_NUM_1** or **FREQ_NUM_2**, the **NRDUCellRimConfig**.***RimRsType1SeqCandNum*** parameter must be set to **SEQ_NUM_1**, **SEQ_NUM_2**, **SEQ_NUM_4**, or **SEQ_NUM_8**.<br>– For a DU cell, if the **NRDUCellRimConfig**.***RimRsFreqCandNum*** parameter is set to **FREQ_NUM_4**, the **NRDUCellRimConfig**.***RimRsType1SeqCandNum*** parameter must be set to **SEQ_NUM_1**, **SEQ_NUM_2**, or **SEQ_NUM_4**.<br>● Scenario with slot assignment 4:1:<br>– For a DU cell, if the **NRDUCellRimConfig**.***RimRsFreqCandNum*** parameter is set to **FREQ_NUM_1**, the **NRDUCellRimConfig**.***RimRsType1SeqCandNum*** parameter must be set to **SEQ_NUM_1**, **SEQ_NUM_2**, **SEQ_NUM_4**, or **SEQ_NUM_8**.<br>– For a DU cell, if the **NRDUCellRimConfig**.***RimRsFreqCandNum*** |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| | | parameter is set to **FREQ_NUM_2**, the **NRDUCellRimConfig**.***RimRsType1SeqCandNum*** parameter must be set to **SEQ_NUM_1**, **SEQ_NUM_2**, or **SEQ_NUM_4**.<br>– For a DU cell, if the **NRDUCellRimConfig**.***RimRsFreqCandNum*** parameter is set to **FREQ_NUM_4**, the **NRDUCellRimConfig**.***RimRsType1SeqCandNum*** parameter must be set to **SEQ_NUM_1** or **SEQ_NUM_2**. |
| RIM-RS Type1 Scrambling ID for IS0 | **NRDUCellRimConfig**.***RimRsType1ScidForIs0*** | If these parameters take valid values, ensure that their values are different within a cell and the same among all of the cells. |
| RIM-RS Type1 Scrambling ID for IS1 | **NRDUCellRimConfig**.***RimRsType1ScidForIs1*** | |
| RIM-RS Type1 Scrambling ID for IS2 | **NRDUCellRimConfig**.***RimRsType1ScidForIs2*** | |
| RIM-RS Type1 Scrambling ID for IS3 | **NRDUCellRimConfig**.***RimRsType1ScidForIs3*** | |
| RIM-RS Type1 Scrambling ID for IS4 | **NRDUCellRimConfig**.***RimRsType1ScidForIs4*** | |
| RIM-RS Type1 Scrambling ID for IS5 | **NRDUCellRimConfig**.***RimRsType1ScidForIs5*** | |
| RIM-RS Type1 Scrambling ID for IS6 | **NRDUCellRimConfig**.***RimRsType1ScidForIs6*** | |
| RIM-RS Type1 Scrambling ID for IS7 | **NRDUCellRimConfig**.***RimRsType1ScidForIs7*** | |

NOTE

- The **NRDUCellRimConfig**.***RimRsType1SetIdSize*** and **NRDUCellRimConfig**.***RimRsType1SetId*** parameter settings affect the success rate and duration of RIM-RS detection. The parameter settings are determined based on the network plan according to the number of IDs required on the live network.
- When the **NRDUCellRimConfig**.***RimRsType1SetIdSize*** parameter is set to a value greater than **20**, the total duration for RIM-RS sending is excessively long. In this case, all gNodeBs in an area need to be planned in a scattered manner and the **NRDUCellRimConfig**.***RimRsType1SetId*** parameter needs to be configured to ensure that there are gNodeBs that send RIM-RSs in each time range in the area. This increases the success rate of remote interference detection and ensures that remote interference management takes effect in time.

**Table 4-9** Parameters used for optimization

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM-RS Function Switch | **NRDUCellRimConfig**.***RimRsFunctionSwitch*** | It is recommended that the **RIM_RS1_NEAR_FAR_SW** option be selected if interference sources from a longer distance need to be detected. |
| RIM-RS Scramble Timer Multiplier Factor | **NRDUCellRimConfig**.***RimRsScrTimerMultiFactor*** | Use the recommended value. |
| RIM-RS Scramble Timer Offset | **NRDUCellRimConfig**.***RimRsScrTimerOffset*** | Use the recommended value. |
| RIM-RS Start Frequency Offset | **NRDUCellRimConfig**.***RimRsStartFreqOffset*** | Use the recommended value. For details about the configuration principles, see **Type-1 RIM-RS**. |
| RIM-RS Start Frequency RE Offset | **NRDUCellRimConfig**.***RimRsStartFreqReOffset*** | Use the recommended value. For details about the configuration principles, see **Type-1 RIM-RS**. |
| UL Pwr Difference Thld | **NRDUCellRimConfig**.***UlPwrDiffThld*** | Set this parameter based on network conditions.<br>You are advised to set this parameter to **-5** if continuous RIM-RS transmission is required for a cell, and **5** if RIM-RSs need to be sent when a cell suffers interference. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM-RS Type1 Intrf Thld | **NRDUCellRimConfig**.***RimRsType1IntrfThld*** | It is recommended that this parameter be set to **-102** when the **NRDUCellRimConfig**.***IntrfAvoidDetectPrdNum*** parameter is set to **0** or the **FORCED_RIM_AVOID_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter is selected, and to **-120** in other cases.<br>Set the **NRDUCellRimConfig**.***RimRsType1IntrfThld*** parameter to **-50** when Hyper Cell, Cell Combination, or High-speed Railway Superior Experience is also used. |
| RIM-RS Type1 Intrf Thld Offset | **NRDUCellRimConfig**.***RimRsType1IntrfThldOfs*** | Use the recommended value. |
| RIM-RS Function Switch | **NRDUCellRimConfig**.***RimRsFunctionSwitch*** | Select the **RIM_RS1_ENOUGH_IND_SW** option when the function of enough/not enough indication for type-1 RIM-RS is required. In other scenarios, you are advised not to select this option. |
| Interfere Avoid Detect Period Num | **NRDUCellRimConfig**.***IntrfAvoidDetectPrdNum*** | ● Value **0** or **1** is recommended when **NRDUCellRimConfig**.***RimRsType2SetIdSize*** is set to **20** or a larger value.<br>● Value **0** is recommended for extremely high interference scenarios where RIM-RSs cannot be detected.<br>● Do not set the **NRDUCellRimConfig**.***IntrfAvoidDetectPrdNum*** parameter to **0** when Hyper Cell, Cell Combination, or High-speed Railway Superior Experience is also used.<br>● Use the recommended value **3** in other scenarios. |
| Interfere Avoid Restore Detect Period Num | **NRDUCellRimConfig**.***IntrfAvoidRstrDetectPrdNum*** | Use the recommended value. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| Interference Avoid UL Pwr Difference Thld | **NRDUCellRimConfig**.***IntrfAvoidUlPwrDiffThld*** | Use the recommended value. |

- Type-2 RIM-RS

  **Table 4-10** and **Table 4-11** describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment.

**Table 4-10** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RIM-RS Function Switch | **NRDUCellRimConfig**.***RimRsFunctionSwitch*** | To enable type-2 RIM-RS, select the **RIM_RS2_SW** option. You are advised not to select this option in other scenarios. |
| RIM-RS Type2 Set ID Size | **NRDUCellRimConfig**.***RimRsType2SetIdSize*** | Set this parameter based on the number of cells for RIM-RS detection. The value of this parameter must be the same for all cells. |
| RIM-RS Type2 Set ID | **NRDUCellRimConfig**.***RimRsType2SetId*** | It is recommended that this parameter be set to different values for cells where this function is enabled. |
| RIM-RS Type2 Sequence Candidate Number | **NRDUCellRimConfig**.***RimRsType2SeqCandNum*** | Use the recommended value. Ensure that the parameter setting is the same among all of the cells. |
| RIM-RS Type2 Scrambling ID for IS0 | **NRDUCellRimConfig**.***RimRsType2ScidForIs0*** | If these parameters take valid values, ensure that their values are different within a cell and the same among all of the cells. |
| RIM-RS Type2 Scrambling ID for IS1 | **NRDUCellRimConfig**.***RimRsType2ScidForIs1*** | |
| RIM-RS Type2 Scrambling ID for IS2 | **NRDUCellRimConfig**.***RimRsType2ScidForIs2*** | |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM-RS Type2 Scrambling ID for IS3 | **NRDUCellRimConfig**.***RimRsType2ScidForIs3*** | |
| RIM-RS Type2 Scrambling ID for IS4 | **NRDUCellRimConfig**.***RimRsType2ScidForIs4*** | |
| RIM-RS Type2 Scrambling ID for IS5 | **NRDUCellRimConfig**.***RimRsType2ScidForIs5*** | |
| RIM-RS Type2 Scrambling ID for IS6 | **NRDUCellRimConfig**.***RimRsType2ScidForIs6*** | |
| RIM-RS Type2 Scrambling ID for IS7 | **NRDUCellRimConfig**.***RimRsType2ScidForIs7*** | |

**Table 4-11** Parameters used for optimization

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM-RS Function Switch | **NRDUCellRimConfig**.***RimRsFunctionSwitch*** | It is recommended that the **RIM_RS2_NEAR_FAR_SW** option be selected if interference sources from a longer distance need to be detected. |

- Remote interference adaptive avoidance

  **Table 4-12** and **Table 4-13** describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment.

**Table 4-12** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM Algorithm Switch | **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** | Select the **RIM_ADAPT_AVOID_SW** option. |

**Table 4-13** Parameters used for optimization

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM Policy Switch | **NRDUCellRimConfig**.*RimPolicySwitch* | It is recommended that the **PWR_COMPENSATE_SW** option be selected if the broadcast channel coverage needs to be retained after remote interference adaptive avoidance is enabled. |
| RIM Policy Switch | **NRDUCellRimConfig**.*RimPolicySwitch* | It is recommended that the **FORCED_RIM_AVOID_SW** option be selected to forcibly retain the adjusted GP after remote interference adaptive avoidance is enabled.<br>Do not select the **FORCED_RIM_AVOID_SW** option when Hyper Cell, Cell Combination, or High-speed Railway Superior Experience is also used. |
| Interference Avoid DL Extension Level | **NRDUCellRimConfig**.*IntrfAvoidDlExtensionLevel* | Use the recommended value **DEFAULT**. Set this parameter to a proper value only when the distance from the interference source to the victim is determined. |
| Channel Calibration Policy Switch | **gNodeBAlgo**.*ChnCalibPolSw* | It is recommended that the **ADAPTIVE_ADJ_SW** option be selected when channel calibration fails due to remote interference. |

- Enhanced remote interference adaptive avoidance

  **Table 4-14** and **Table 4-15** describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment.

**Table 4-14** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM Algorithm Switch | **NRDUCellAlgoSwitch**.*RimAlgoSwitch* | Select the **RIM_ADAPT_AVOID_ENH_SW** option. |

**Table 4-15** Parameters used for optimization

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| Channel Calibration Policy Switch | **gNodeBAlgo.***ChnCalibPolSw* | It is recommended that the **ADAPTIVE_ADJ_SW** option be selected when channel calibration fails due to remote interference. |

### 4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

- Type-1 RIM-RS
```
//Setting parameters related to type-1 RIM-RS
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimRsType1SetIdSize=20, RimRsType1SetId=1023,
RimRsFreqCandNum=FREQ_NUM_1, RimRsType1SeqCandNum=SEQ_NUM_2,
RimRsType1ScidForIs0=0, RimRsType1ScidForIs1=10;
```
- Remote interference adaptive avoidance
```
//RIM-RS must be configured in advance.
//Enabling remote interference adaptive avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, RimAlgoSwitch=RIM_ADAPT_AVOID_SW-1;
```
- Enhanced remote interference adaptive avoidance
```
//RIM-RS must be configured in advance.
//Enabling enhanced remote interference adaptive avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, RimAlgoSwitch=RIM_ADAPT_AVOID_ENH_SW-1;
```
- Type-2 RIM-RS
```
//Setting parameters related to type-2 RIM-RS
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimRsType2SetIdSize=20, RimRsType2SetId=1023,
RimRsFreqCandNum=FREQ_NUM_1, RimRsType2SeqCandNum=SEQ_NUM_2,
RimRsType2ScidForIs0=1, RimRsType2ScidForIs1=11;
//Turning on the type-2 RIM-RS function switch
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimRsFunctionSwitch=RIM_RS2_SW-1;
```

## Optimization Command Examples

```
//Enabling near-far functionality for type-1 RIM-RS
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimRsFunctionSwitch=RIM_RS1_NEAR_FAR_SW-1;
//Enabling the function of enough/not enough indication for type-1 RIM-RS
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimRsFunctionSwitch=RIM_RS1_ENOUGH_IND_SW-1;
//Enabling near-far functionality for type-2 RIM-RS
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimRsFunctionSwitch=RIM_RS2_NEAR_FAR_SW-1;
//Configuring the start frequency-domain offset for RIM-RS
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimRsStartFreqOffset=0, RimRsStartFreqReOffset=0;
//Configuring the time-domain scrambling factor for RIM-RS
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimRsScrTimerMultiFactor=0, RimRsScrTimerOffset=0;
```

```
//Setting thresholds (In this example, the NRDUCellRimConfig.RimRsType1IntrfThld parameter is set to
-120.)
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, UlPwrDiffThld=5, RimRsType1IntrfThld=-120,
RimRsType1IntrfThldOfs=-5, IntrfAvoidUlPwrDiffThld=10;
//Reconfiguring the IntrfAvoidDetectPrdNum and IntrfAvoidRstrDetectPrdNum parameters
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, IntrfAvoidDetectPrdNum=3,IntrfAvoidRstrDetectPrdNum=15;
//Modifying the GP adjustment level for remote interference adaptive avoidance
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, IntrfAvoidDlExtensionLevel=DEFAULT;
//Enabling broadcast channel power compensation
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimPolicySwitch=PWR_COMPENSATE_SW-1;
//(Recommended when the adjusted GP state needs to be forcibly maintained) Selecting the
FORCED_RIM_AVOID_SW option
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimPolicySwitch=FORCED_RIM_AVOID_SW-1;
//(Recommended when channel calibration fails due to remote interference) Selecting the
ADAPTIVE_ADJ_SW option of the ChnCalibPolSw parameter
MOD GNODEBALGO: ChnCalibPolSw=ADAPTIVE_ADJ_SW-1;
```

**NOTE**

- It is recommended that **NRDUCellRimConfig**.***UlPwrDiffThld*** be set to a relatively small value so that the gNodeB is more likely to proactively send the RIM-RS. In this way, effective interference avoidance can be implemented after an interfering cell detects the RIM-RS.
- Before selecting the **PWR_COMPENSATE_SW** or **FORCED_RIM_AVOID_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter, ensure that the **RIM_ADAPT_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter is selected.

### Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling broadcast channel power compensation
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimPolicySwitch=PWR_COMPENSATE_SW-0;
//Disabling forcible GP adjustment
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimPolicySwitch=FORCED_RIM_AVOID_SW-0;
//Turning off the type-2 RIM-RS function switch
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimRsFunctionSwitch=RIM_RS2_SW-0;
//Disabling remote interference adaptive avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, RimAlgoSwitch=RIM_ADAPT_AVOID_SW-0;
//Disabling enhanced remote interference adaptive avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, RimAlgoSwitch=RIM_ADAPT_AVOID_ENH_SW-0;
```

### 4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 4.4.2 Activation Verification

### Using MML Commands

- Run the **LST NRDUCELLALGOSWITCH** command. If any of the following is displayed in the command output, the feature switch has been turned on.

```
RIM Algorithm Switch  =  RIM Adaptation Avoidance Switch:On
RIM Algorithm Switch  =  RIM Adaptation Avoidance Enh Switch:On
```

- Run the **DSP NRDUCELLRIMCONFIG** command. If any of the following is displayed in the command output, this feature has taken effect, in which case the gNodeB sends RIM-RSs but does not perform GP expansion (through remote interference adaptive avoidance) or dynamically adjust the time domain range for scheduling UEs (through enhanced remote interference adaptive avoidance).
```
RIM Adaptation Avoidance State = Send Only RIM-RS
Enhanced RIM Adaptation Avoidance State = Send Only RIM-RS
```
- Run the **DSP NRDUCELLRIMCONFIG** command. If the following information is displayed in the command output, the remote interference adaptive avoidance function has taken effect.
```
RIM Adaptation Avoidance State  =  GP Expanded
```
- Run the **DSP NRDUCELLRIMCONFIG** command. If any of the following is displayed in the command output, the enhanced remote interference adaptive avoidance function has taken effect.
```
Enhanced RIM Adaptation Avoidance State  =  RIM Light Coordinated Scheduling
Enhanced RIM Adaptation Avoidance State  =  RIM Heavy Coordinated Scheduling
```

## Using Counters

Check whether the functions take effect based on indicator changes. For details, see the following table and **4.4.3 Network Monitoring**.

| Counter Name | Counter Description | Description |
|---|---|---|
| **N.Rim.AdaptAvoid.Dur** | Effective duration of remote interference adaptive avoidance | The counter measures the duration in which a cell is in the GAP expansion state to avoid remote interference when remote interference adaptive avoidance is enabled. |
| **N.Rim.EnhacedAdaptAvoid.LightCoordinated.Dur** | Effective duration of light coordinated scheduling of enhanced remote interference adaptive avoidance | The counter measures the duration in which a cell is in the light coordinated scheduling state to avoid remote interference when enhanced remote interference adaptive avoidance is enabled. |
| **N.Rim.EnhacedAdaptAvoid.HeavyCoordinated.Dur** | Effective duration of heavy coordinated scheduling of enhanced remote interference adaptive avoidance | The counter measures the duration in which a cell is in the heavy coordinated scheduling state to avoid remote interference when enhanced remote interference adaptive avoidance is enabled. |

### 4.4.3 Network Monitoring

Observe whether the following indicators change as expected:

- **Call Setup Success Rate (CU)**, which should increase
- **Intra-Frequency Handover Out Success Rate (CU)**, which should increase
- **User Uplink Average Throughput (DU)**, which should increase
- **Cell Uplink Average Throughput (DU)**, which should increase
- **N.GAP.LastSymbol.Pwr** counter value minus the **N.UL.Last.Symbol13.Pwr** counter value, which should decrease
- **N.UL.NI.Avg**, which should decrease

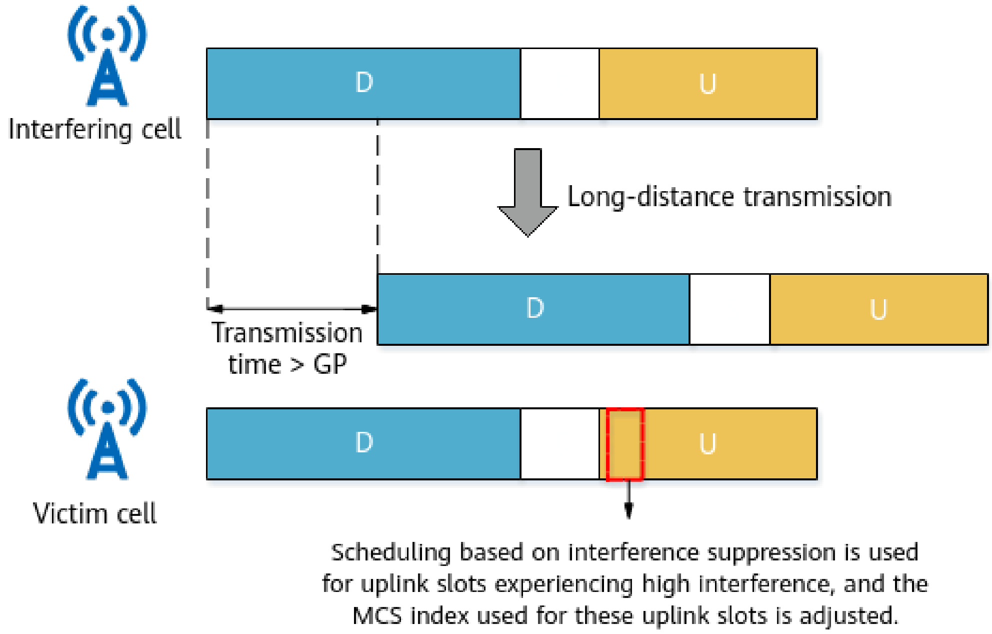
# 5 Adaptive Interference Suppression

## 5.1 Remote Interference Adaptive Suppression

### 5.1.1 Principles

Remote interference adaptive suppression mitigates the decrease in the average uplink cell throughput caused by remote interference. This function is controlled by the **RIM_ADAPT_SUPPRESSION_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter. After this function is enabled, the gNodeB periodically detects interference to different uplink slots. If the interference strength difference between symbols in multiple uplink slots reaches the value of the **NRDUCellRimConfig**.***UlSlotPwrDiffThld*** parameter, the gNodeB dynamically adjusts the scheduling policies for different uplink slots based on the detection result. As shown in **Figure 5-1**, **scheduling based on interference suppression is used for uplink slots experiencing high interference, and the MCS index used for these uplink slots is decreased** by an amount specified by **NRDUCellRimConfig**.***IntrfUlSlotInitMcsOffset*** to reduce the impact of remote interference.

**Figure 5-1** Remote interference adaptive suppression




NOTE

- According to slope characteristics of the atmospheric duct, the closer an uplink slot to the GP, the more severe the interference to the slot.
- For slots subjected to interference, the larger the initial MCS index, the more the bit errors, and the more significant the decrease in the throughput.

## 5.1.2 Network Analysis

### 5.1.2.1 Benefits

#### Deployment Suggestions

It is recommended that this function be enabled when mutual RIM-RS detection is not supported between the gNodeB serving the interfering cells and the gNodeB serving the victim cells. In such scenarios, remote interference exists between the NR system and other systems, or mutual RIM-RS detection between NR systems is not supported because the overlapping bandwidth is less than 20 MHz.

It is recommended that remote interference adaptive suppression be enabled when the uplink throughput of a cell decreases in certain periods and all of the following conditions are met:

- The measured power of different symbols in the first uplink slot following the self-contained slot exhibits the slope characteristics. Specifically, the values of **N.UL.First.Symbol0.Pwr**, **N.UL.First.Symbol2.Pwr**, **N.UL.First.Symbol4.Pwr**, and **N.UL.First.Symbol6.Pwr** decrease in sequence.
- The level of interference to the first uplink slot following the self-contained slot meets the following condition:

  **N.UL.First.Symbol0.Pwr – N.UL.First.Symbol6.Pwr** > 10 dB

Enabling this function when the preceding conditions are not met may decrease the average uplink cell throughput.

NOTE

It is recommended that the functions described in **4 Remote Interference Management** be enabled when mutual RIM-RS detection is available between the gNodeB serving the interfering cells and the gNodeB serving the victim cells.

### Benefits

Enabling remote interference adaptive suppression enhances the gNodeB's capabilities of resisting and suppressing remote interference. The MCS indexes used for the uplink slots subjected to interference decrease, and those for interference-free uplink slots increase. The overall uplink MCS index of the network increases, the uplink cell IBLER decreases, and the average uplink cell throughput increases. For details about indicators involved, see **5.1.4.3 Network Monitoring**.

### 5.1.2.2 Impacts

The impacts between this function and other functions are described in **5.1.3.2.3 Function Impacts**.

There is no impact on the network.

## 5.1.3 Requirements

### 5.1.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
| --- | --- | --- | --- |
| FOFD-061220 | Adaptive Interference Suppression | NR0S00ATFS00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

### 5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 5.1.3.2.1 Prerequisite Functions

None

#### 5.1.3.2.2 Mutually Exclusive Functions

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Intra-Frequency Assistant Cell | **NRDUCell**.***SupplementaryCellInd*** set to **INTRA_FREQ_ASSISTANT_CELL** | None | When the **NRDUCell**.***SupplementaryCellInd*** parameter is set to **INTRA_FREQ_ASSISTANT_CELL**, all these options must be deselected: (1) **RIM_ADAPT_AVOID_SW**, **RIM_ADAPT_AVOID_ENH_SW**, and **RIM_ADAPT_SUPPRESSION_SW** of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter, and (2) **INTRA_GNB_SU_COMM_SW** and **INTER_CELL_INTRF_AVOID_SW** of the **NRDuCellCollabserv**.***MultiCellMimoSwitch*** parameter. |

#### 5.1.3.2.3 Function Impacts

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| NR low power consumption mode | **LOW_PWR_MODE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Green Site* | When both low power consumption mode and remote interference adaptive suppression are enabled and a customized backup power saving policy is used in low power consumption mode, the following rules apply:<br>● If the coverage compensation policy in NR low power consumption mode is disabled (by deselecting the **COV_COMPENSATION_POLICY_SW** option of the **NRDUCellPowerSaving**.***LowPwrModeAlgoSwitch*** parameter), channel shutdown in low power consumption mode does not take effect.<br>● If the coverage compensation policy in NR low power consumption mode is enabled, low power consumption mode and remote interference adaptive suppression can both take effect. In this case, coverage performance of low power |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| | | | consumption mode may deteriorate. |
| Intelligent uplink multi-layer SINR prediction | **UL_SU_SINR_INTEL_PREDICT_SW** option of the **NRDUCellAiAlgo**.***AiAmcAlgoSwitch*** parameter | *MIMO (TDD)* | After remote interference adaptive suppression takes effect, the proportion of occasions where intelligent uplink multi-layer SINR prediction takes effect will be affected and accordingly the gains of intelligent uplink multi-layer SINR prediction may be lowered. |
| High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED** | *High Speed Mobility* | In high-speed scenarios, remote interference adaptive suppression yields fewer gains than it would in low-speed scenarios. |

### 5.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

All NR TDD-capable 2T2R, 4T4R, 8T8R, 32T32R, and 64T64R RF modules support this function.

#### 5.1.3.4 Cells

The SCS is 30 kHz.

Remote interference adaptive suppression supports only the following slot configurations:

- Single-period 8:2
- Dual-period 8:2

#### 5.1.3.5 Others

None

### 5.1.4 Operation and Maintenance

#### 5.1.4.1 Data Configuration

##### 5.1.4.1.1 Data Preparation

Table 5-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 5-1** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM Algorithm Switch | **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** | Select the **RIM_ADAPT_SUPPRESSION_SW** option. |
| Uplink Slot Power Difference Threshold | **NRDUCellRimConfig**.***UlSlotPwrDiffThld*** | Set this parameter based on the interference levels of uplink slots on the live network. It is recommended that this parameter be set to a value greater than **10**. Setting this parameter to an overly small value makes it more likely to trigger MCS index adjustment in low interference scenarios, leading to deterioration in uplink performance. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| Interfered Uplink Slot Initial MCS Offset | **NRDUCellRimConfig**.***IntrfUlSlotInitMcsOffset*** | Set this parameter based on the level of interference to uplink slots on the live network. You are advised to set this parameter to **1** when the interference level of the first uplink slot following the self-contained slot reaches 10 dB to 15 dB, and to **2** when the interference level reaches above 15 dB. Setting this parameter to an overly large value leads to relatively low MCS indexes used for uplink slots, leading to deterioration in uplink performance. |

### 5.1.4.1.2 Using MML Commands

Before using MML commands, refer to **5.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Turning on the remote interference adaptive suppression switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, RimAlgoSwitch=RIM_ADAPT_SUPPRESSION_SW-1;

//Configuring the adaptive triggering threshold and MCS index reduction offset for remote interference adaptive suppression
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, UlSlotPwrDiffThld=10, IntrfUlSlotInitMcsOffset=1;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the remote interference adaptive suppression switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, RimAlgoSwitch=RIM_ADAPT_SUPPRESSION_SW-0;
```

### 5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

#### 5.1.4.2 Activation Verification

Method 1

**Step 1** Run the **LST NRDUCELLALGOSWITCH** command. If the following information is displayed in the command output, this feature has been enabled.

```
RIM Algorithm Switch  =  RIM Adaptation Suppression Switch:On
```

**Step 2** Run the **DSP NRDUCELLRIMCONFIG** command. If the following information is displayed in the command output, the remote interference adaptive suppression function has taken effect.

```
RIM Adaptation Suppress State  =  RIM Suppression Scheduling
```

**----End**

Method 2

Subscribe to the **PERIOD_CELL_RIM_INFO** event. If the value of **TddRimUlStateFlag** is **1**, remote interference adaptive suppression has taken effect.

#### 5.1.4.3 Network Monitoring

Observe whether the following indicators change as expected:

- **User Uplink Average Throughput (DU)**, which should increase
- **Cell Uplink Average Throughput (DU)**, which should increase

## 5.2 Remote Interference Resistance for PUSCH Demodulation

### 5.2.1 Principles

According to the slope characteristics of remote interference, a symbol closer to the GP suffers stronger interference. As such, the interference characteristics of the front-loaded DMRS and auxiliary DMRS of the PUSCH are different. If the front-loaded DMRS and auxiliary DMRS are used for channel estimation, the PUSCH demodulation performance will be affected. Remote interference resistance for PUSCH demodulation is controlled by the **RIM_PUSCH_DEMOD_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter. After this function is enabled, the gNodeB **adjusts the channel estimation result during PUSCH demodulation to reduce the weight of interference data involved in demodulation, enhancing PUSCH demodulation performance.**

### 5.2.2 Network Analysis

### 5.2.2.1 Benefits

#### Deployment Suggestions

It is recommended that this function be enabled when mutual RIM-RS detection is not supported between the gNodeB serving the interfering cells and the gNodeB serving the victim cells. In such scenarios, remote interference exists between the NR system and other systems, or mutual RIM-RS detection between NR systems is not supported because the overlapping bandwidth is less than 20 MHz.

Remote interference resistance for the PUSCH demodulation is recommended when the uplink throughput decreases due to atmospheric duct interference in a cell and all of the following conditions are met:

- The measured power of different symbols in the first uplink slot following the self-contained slot exhibits the slope characteristics. Specifically, the values of **N.UL.First.Symbol0.Pwr**, **N.UL.First.Symbol2.Pwr**, **N.UL.First.Symbol4.Pwr**, and **N.UL.First.Symbol6.Pwr** decrease in sequence.
- The level of interference to the first uplink slot following the self-contained slot meets the following condition: **N.UL.First.Symbol0.Pwr** – **N.UL.First.Symbol6.Pwr** > 15 dB.

If this function is enabled but not all preceding conditions are met, the uplink cell throughput may decrease.

#### Benefits

Enabling remote interference resistance for PUSCH demodulation enhances the gNodeB's capabilities of resisting and suppressing remote interference. When remote interference affects uplink slots and the strength of interference to the first uplink slot following the self-contained slot reaches 15 dB to 25 dB, the average uplink cell throughput and average uplink UE throughput increase significantly. For details about the performance indicators involved, see **5.2.4.3 Network Monitoring**.

### 5.2.2.2 Impacts

The impacts between this function and other functions are described in **5.2.3.2.3 Function Impacts**.

There is no impact on the network.

## 5.2.3 Requirements

### 5.2.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-061220 | Adaptive Interference Suppression | NR0S00ATFS00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

### 5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 5.2.3.2.1 Prerequisite Functions

None

#### 5.2.3.2.2 Mutually Exclusive Functions

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | The two functions are mutually exclusive: Hyper Cell and remote interference resistance for PUSCH demodulation. |
| Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** | *Cell Combination* | The two functions are mutually exclusive: Cell Combination and remote interference resistance for PUSCH demodulation. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Uplink DMRS type 2 | **NRDUCellPusch**.***UlDmrsType*** set to **TYPE2** | None | When remote interference resistance for PUSCH demodulation is enabled, **NRDUCellPusch**.***UlDmrsType*** cannot be set to **TYPE2**. |
| High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED** | *High Speed Mobility* | The two functions are mutually exclusive: High-speed Railway Superior Experience and remote interference resistance for PUSCH demodulation. |

#### 5.2.3.2.3 Function Impacts

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| NR low power consumption mode | **LOW_PWR_MODE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Green Site* | When both low power consumption mode and remote interference resistance for PUSCH demodulation are enabled and a customized backup power saving policy is used in low power consumption mode, the following rules apply:<br>● If the coverage compensation policy in NR low power consumption mode is disabled (by deselecting the **COV_COMPENSATION_POLICY_SW** option of the **NRDUCellPowerSaving**.***LowPwrModeAlgoSwitch*** parameter), channel shutdown in low power consumption mode does not take effect.<br>● If the coverage compensation policy in NR low power consumption mode is enabled, low power consumption mode and remote interference resistance for PUSCH demodulation can both take effect. In this case, coverage |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| | | | performance of low power consumption mode may deteriorate. |
| Time-domain adaptive filtering | **PUSCH_TIME_DOM_ADAPT_FILTER_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter | *Turbo Uplink (Low-Frequency TDD)* | When remote interference resistance for PUSCH demodulation takes effect for UEs, time-domain adaptive filtering does not take effect for these UEs. |
| Intelligent uplink multi-layer SINR prediction | **UL_SU_SINR_INTEL_PREDICT_SW** option of the **NRDUCellAiAlgo**.***AiAmcAlgoSwitch*** parameter | *MIMO (TDD)* | Remote interference resistance for PUSCH demodulation adversely affects the proportion of occasions where intelligent uplink multi-layer SINR prediction takes effect. As a result, intelligent uplink multi-layer SINR prediction may deliver fewer benefits. |

### 5.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

### 5.2.3.4 Cells

The SCS is 30 kHz.

### 5.2.3.5 Others

None

## 5.2.4 Operation and Maintenance

### 5.2.4.1 Data Configuration

#### 5.2.4.1.1 Data Preparation

**Table 5-2** describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 5-2** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM Algorithm Switch | **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** | Select the **RIM_PUSCH_DEMOD_SW** option. |

#### 5.2.4.1.2 Using MML Commands

Before using MML commands, refer to **5.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

### Activation Command Examples

```
//Enabling remote interference resistance for PUSCH demodulation
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,RimAlgoSwitch=RIM_PUSCH_DEMOD_SW-1;
```

#### Deactivation Command Examples

```
//Disabling remote interference resistance for PUSCH demodulation
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,RimAlgoSwitch=RIM_PUSCH_DEMOD_SW-0;
```

#### 5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

> **NOTE**
>
> Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 5.2.4.2 Activation Verification

Subscribe to the **PERIOD_CELL_RIM_INFO** event. If the value of **TddRimUlStateFlag** is **2** or **3**, remote interference resistance for PUSCH demodulation has taken effect.

### 5.2.4.3 Network Monitoring

Check whether the **Cell Uplink Average Throughput (DU)** increases.

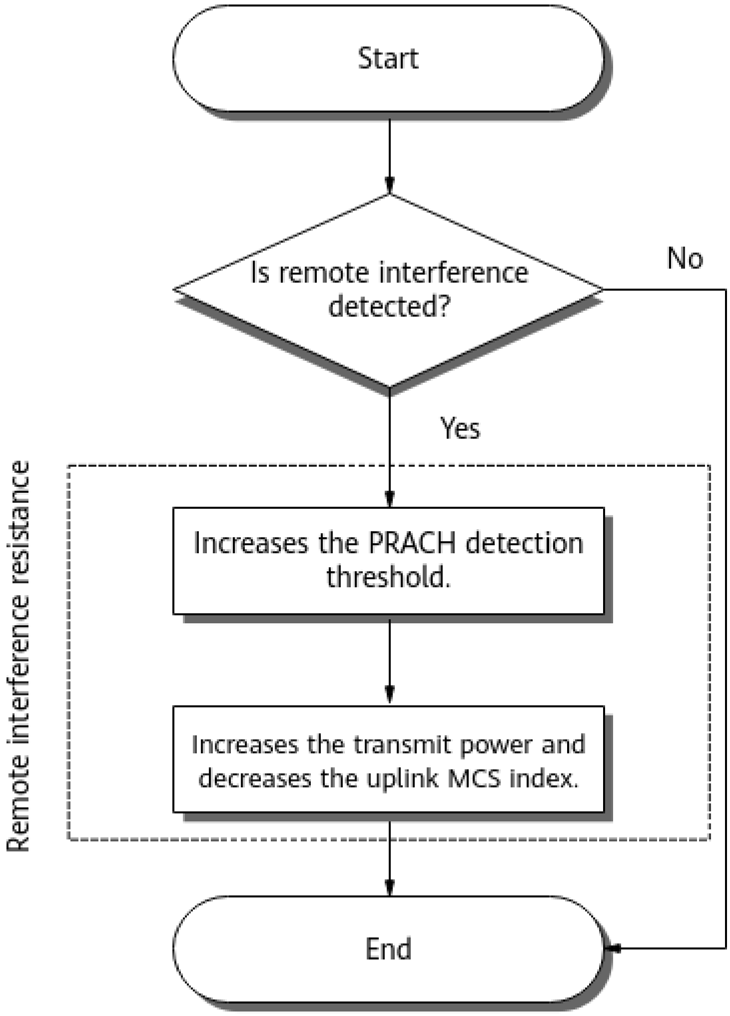
Check whether the **User Uplink Average Throughput (DU)** increases.

# 5.3 Remote Interference Resistance for Random Access

## 5.3.1 Principles

When a cell suffers remote interference, UE access to this cell will be affected due to interference to uplink slots. Huawei introduces the function of remote interference resistance for random access to enhance signaling reliability and increase the random access success rate of a cell. This function is controlled by the **RA_REMOTE_INTRF_RESIST_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter. **Figure 5-2** shows how this function works.

**Figure 5-2** Working process of remote interference resistance for random access




1. Performs remote interference detection.

   The gNodeB performs remote interference detection as follows:

   – The gNodeB determines that a cell suffers remote interference if the following conditions are met within three consecutive minutes: The strength of interference to the uplink symbols gradually decreases as the distance between an uplink symbol and GP increases (meeting the slope characteristic of atmospheric duct interference); the interference strength difference between the uplink symbols is greater than the **NRDUCellRimConfig**.***UlPwrDiffThld*** parameter value. In this case, the gNodeB performs remote interference resistance.

   – If the preceding conditions are not met within three consecutive minutes after remote interference resistance takes effect, the gNodeB terminates remote interference resistance and restores the original configuration.

2. Increases the PRACH detection threshold.

   The PRACH detection threshold for UEs that perform contention-based random access to cells is fixedly increased by 300% to control the access of cell edge users (CEUs). This step does not affect UEs that perform non-contention-based random access to cells.

3. Increases the transmit power and decreases the uplink MCS index.
   – In the random access phase, the uplink MCS index fixedly decreases by 2.
   – When the **RA_POWER_CONTROL_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter is selected, the gNodeB increases the UE transmit power in the random access phase and does not respond to power decrease instructions. Increasing the transmit power improves the reliability in signal reception. When this option is deselected, the transmit power is not increased.

NOTE

The **RA_POWER_CONTROL_SW** option of the **NRDUCellRimConfig**.***RimPolicySwitch*** parameter can be selected only when the **RA_REMOTE_INTRF_RESIST_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter is selected.

# 5.3.2 Network Analysis

## 5.3.2.1 Benefits

### Deployment Suggestions

It is recommended that this function be enabled in cells for which the values of indicators related to the random access success rate need to be increased only in scenarios where remote interference exists and cannot be eliminated from the interference source side due to objective reasons (for example, in scenarios where cells are deployed across different countries or operators).

### Benefits

When the remote interference strength is in the range of 20 dB to 30 dB, the random access success rate increases by about 2%, and the RRC connection setup success rate may increase. For details about the performance indicators involved, see **5.3.4.3 Network Monitoring**.

## 5.3.2.2 Impacts

The impacts between this function and other functions are described in **5.3.3.2.3 Function Impacts**.

- Increasing the PRACH detection threshold restricts the access of CEUs, affecting user experience. The CEUs attempt to access the network by sending preambles multiple times. If the time-frequency domain positions of PRACHs between cells are not aligned, a large number of PRACHs will cause interference to the PUSCHs/PUCCHs of neighboring cells, increasing the uplink BLERs of neighboring cells.

  Uplink initial transmission block error rate = (**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**)

- Increasing the UE transmit power increases inter-cell interference.
- Decreasing the uplink MCS index may increase the uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) of the cell.

## 5.3.3 Requirements

### 5.3.3.1 Licenses

There are no license requirements for basic functions.

### 5.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 5.3.3.2.1 Prerequisite Functions

None

#### 5.3.3.2.2 Mutually Exclusive Functions

None

#### 5.3.3.2.3 Function Impacts

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| NR low power consumption mode | **LOW_PWR_MODE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Green Site* | When both low power consumption mode and remote interference resistance for random access are enabled and a customized backup power saving policy is used in low power consumption mode, the following rules apply:<br>● If the coverage compensation policy in NR low power consumption mode is disabled (by deselecting the **COV_COMPENSATION_POLICY_SW** option of the **NRDUCellPowerSaving**.***LowPwrModeAlgoSwitch*** parameter), channel shutdown in low power consumption mode does not take effect.<br>● If the coverage compensation policy in NR low power consumption mode is enabled, low power consumption mode and remote interference resistance for random access can both take effect. In this case, coverage performance of low power consumption mode may deteriorate. |
| Extended Cell Range (TDD) | **NRDUCell**.***CellRadius*** set to a value greater than **14500** | *Extended Cell Range* | Remote interference resistance for random access affects the access of CEUs. As a result, CEUs may fail to access the network in Extended Cell Range scenarios. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Extended Cell Range of 90 km (TDD) | **NRDUCell**.***CellRadius*** set to a value greater than **60000** and less than or equal to **90000** | *Extended Cell Range* | Remote interference resistance for random access affects the access of CEUs. As a result, CEUs may fail to access the network in Extended Cell Range of 90 km scenarios. |
| High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED** | *High Speed Mobility* | In high-speed scenarios, remote interference resistance for random access yields fewer gains than it would in low-speed scenarios. |
| Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | Remote interference resistance for random access takes effect at the cell level. In a cell with multiple TRPs, if interference that is present at a TRP meets the conditions for triggering this function, this function will be triggered for that TRP and for other TRPs, even though the other TRPs do not experience interference. Consequently, the number of UEs served by the other TRPs may decrease, accompanied by fluctuations in the values of access-related counters. |
| Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** | *Cell Combination* | Remote interference resistance for random access takes effect at the cell level. In a cell with multiple TRPs, if interference that is present at a TRP meets the conditions for triggering this function, this function will be triggered for that TRP and for other TRPs, even though the other TRPs do not experience interference. Consequently, the number of UEs served by the other TRPs may decrease, accompanied by fluctuations in the values of access-related counters. |

#### 5.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

All NR TDD-capable low-frequency RF modules that work in bands n38, n40, n41, n77, n78, and n79 support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

#### 5.3.3.4 Cells

The SCS is 30 kHz.

#### 5.3.3.5 Others

None

## 5.3.4 Operation and Maintenance

### 5.3.4.1 Data Configuration

#### 5.3.4.1.1 Data Preparation

**Table 5-3** describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 5-3** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RIM Algorithm Switch | **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** | Select the **RA_REMOTE_INTRF_RESIST_SW** option. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RIM Policy Switch | **NRDUCellRimConfig**.***RimPolicySwitch*** | Select the **RA_POWER_CONTROL_SW** option. |
| UL Pwr Difference Thld | **NRDUCellRimConfig**.***UlPwrDiffThld*** | Use the recommended value. |

#### 5.3.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling remote interference resistance for random access
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,RimAlgoSwitch=RA_REMOTE_INTRF_RESIST_SW-1;
//Turning on the switch for random access power control
MOD NRDUCELLRIMCONFIG: NrDuCellId=0,RimPolicySwitch=RA_POWER_CONTROL_SW-1;
//Setting the thresholds
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, UlPwrDiffThld=5;
```

## Deactivation Command Examples

```
//Turning off the switch for random access power control
MOD NRDUCELLRIMCONFIG: NrDuCellId=0,RimPolicySwitch=RA_POWER_CONTROL_SW-0;
//Disabling remote interference resistance for random access
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,RimAlgoSwitch=RA_REMOTE_INTRF_RESIST_SW-0;
```

#### 5.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 5.3.4.2 Activation Verification

- Run the **LST NRDUCELLALGOSWITCH** command. If the following information is displayed in the command output, this function has been enabled.
```
RIM Algorithm Switch  =  RA Remote Interference Resistance Sw:On
```
- Observe related indicators. For details, see **5.3.4.3 Network Monitoring**.

#### 5.3.4.3 Network Monitoring

Observe whether the following indicators change as expected:

- Contention-based random access success rate, which should increase. The success rate is calculated using the following formula: (**N.RA.Contention.Resolution.Succ**/**N.RA.Contention.Att**) x 100%.
- **RRC Setup Success Rate (CU)**, which should increase

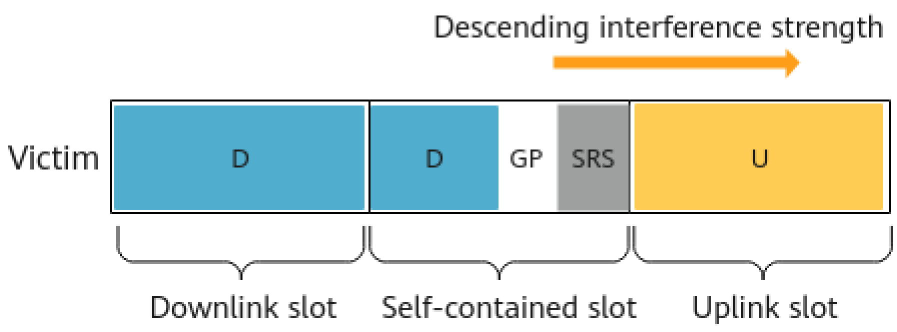
# 6 Proactive Avoidance of Remote Interference

This section describes proactive avoidance of remote interference in low-speed cells.

## 6.1 Principles

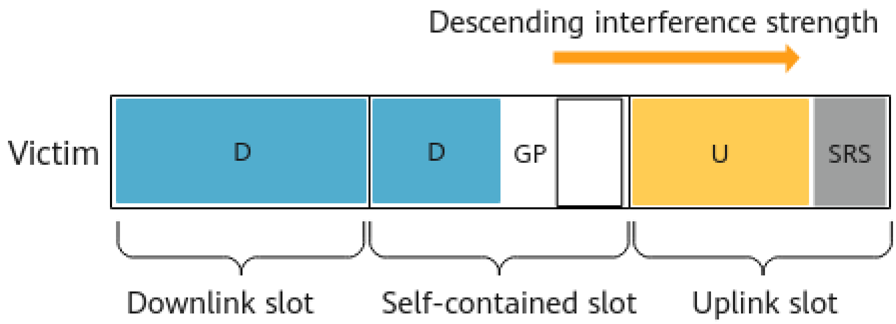
SRSs serve as important channel estimation signals. Their time-domain positions are most susceptible to remote interference if the traditional TDD channel structure (as shown in **Figure 6-1**) is used. This makes a TDD system's performance contingent on remote interference.

**Figure 6-1** Traditional channel structure




**Luckily, proactive avoidance of remote interference is introduced. This function enables the reallocation of SRSs to time-domain positions that experience low uplink interference, as shown in Figure 6-2. This adjustment to the channel structure is based on the sloping character of interference strength, and provides a proactive and flexible approach to interference avoidance**.

**Figure 6-2** Optimized channel structure




Proactive avoidance of remote interference is jointly controlled by the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs.*SrsAlgoExtSwitch*** parameter. After this function is enabled, the gNodeB uses the uplink symbols that experience the lowest remote interference for SRS transmission, and the SRS-specific uplink symbol positions are different for the cells with different PCI mod 3 values.

# 6.2 Network Analysis

## 6.2.1 Benefits

### Deployment Suggestions

It is recommended that proactive avoidance of remote interference be enabled for all gNodeBs that meet the following conditions in areas where atmospheric duct interference occurs:

- Average value of **N.GAP.LastSymbol.Pwr** > –100 dBm
- Average value of **N.S.Symbol10.NI.Avg**, **N.S.Symbol11.NI.Avg**, **N.S.Symbol12.NI.Avg**, and **N.S.Symbol13.NI.Avg** > –100 dBm

When the preceding conditions are not met, enabling this function may decrease the average downlink and uplink throughputs of cells and UEs.

This function requires continuous deployment planning. It cannot be enabled only for some sites in mixed networking. If this function is enabled only for some sites in mixed networking, there will be PUSCH and SRS interference among neighboring cells. As a result, uplink/downlink user experience deteriorates and the indicators related to UE access may deteriorate.

### Benefits

Enabling proactive avoidance of remote interference mitigates the impact of remote interference and increases the **User Downlink Average Throughput (DU)**.

## 6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.2.3 Function Impacts**.

The following impacts arise from selecting the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter:

- SRS demodulation performance improves, which may decrease the service drop rate. If this occurs, the number of CEUs increases or the number of repeated access attempts decreases. Moreover, CEUs camp in the given cell for longer periods, which may decrease reported CQIs and the access success rate.
- If certain symbols of an uplink slot are allocated to SRS, the number of PUSCH symbols decreases. This results in a decline in the performance of demodulating signaling (such as Msg3) from UEs under weak coverage, which in turn reduces the RACH setup success rate and RRC connection setup success rate.
- If both 'LTE TDD and NR dynamic spectrum sharing' and 'proactive avoidance of remote interference' are enabled for an NR cell, mutual interference occurs between the SRS in the NR cell and the PUSCH signal in the involved LTE cell because these signals overlap in the time domain.
- If certain symbols of an uplink slot are allocated to SRS, fewer symbols are available for the PUCCH and PRACH. This worsens demodulation performance and increases the probability of resource allocation conflicts. When many UEs are simultaneously trying to access the network or CEUs are trying to access the network, their access delay may increase. To improve PUCCH demodulation performance and increase resources available for the PUCCH, it is recommended that settings of the following parameters be optimized:
  - **PUCCH_MRC_IRC_SW** option of the **NRDUCellPucch**.***PucchReceiveEnhSwitch*** parameter: You are advised to select this option to enable the PUCCH MRC/IRC adaptation function. This helps improve PUCCH demodulation performance. For details about the PUCCH MRC/IRC adaptation function, see *Channel Management*.
  - **NRDUCellPusch**.***SchMultiSlivTypeUeSwitch***: You are advised to set this parameter to **ON** to enable the function of PUSCH scheduling for multi-SLIV UEs. After this function is enabled, optimal symbols are dynamically selected in self-contained slots for PUSCH scheduling based on UE capabilities.
  - **NRDUCellPusch**.***PuschSharePucchResPol***: You are advised to set this parameter to **NOT_SHARED** to disable PUCCH resource sharing with the PUSCH. This prevents the PUSCH from occupying PUCCH RBs.
- When the positions of SRS symbols are changed, fewer symbols are available for the PUSCH. In weak coverage areas, smaller data blocks are transmitted from CEUs in uplink slots with SRS transmission, which leads to more fragments needed to transmit large signaling messages during network access. Overall, this increases the probability of message transmission failures, which may decrease the QoS flow setup success rate.
- As the number of available PUSCH symbols decreases when SRS uses some of the symbols allocated to the PUSCH, the average uplink UE throughput is negatively affected. For a cell for which (1) the **NRDUCell**.***SlotAssignment***

parameter is set to **8_2_DDDDDDDSUU** (indicating the single-period 8:2 slot configuration), **8_2_DDDSUUDDDD** (indicating the dual-period 8:2 slot configuration), or **4_1_DDDSU** (indicating the 4:1 slot configuration), and (2) the **NRDUCell**.***SlotStructure*** parameter is set to **SS54**, **SS84**, or **SS4**, it is recommended to select the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter, and to select any of the **8_2_SS54_ADAPT_SW**, **8_2_SS84_ADAPT_SW**, and **4_1_SS4_ADAPT_SW** options of the **gNodeBAlgo**.***UlSchSwitch*** parameter depending on the slot configuration and slot structure. Points to consider are as follows:

- The preceding options take effect only on UEs engaged in common data transmission services. This means that the functions controlled by these options do not take effect on UEs using special services (such as VoNR, URLLC, and FWA services).
- The preceding options can take effect only when certain requirements are met. Before selecting them, contact Huawei engineers.
- After the preceding options are selected, the gNodeB comprehensively evaluates specific factors, such as (1) the strength of interference in self-contained slots (which must be less than or equal to the PUSCH interference threshold specified by the **NRDUCellSrs**.***PuschIntrfThld*** parameter), (2) uplink load, and (3) traffic model. Based on the result of evaluation, the gNodeB optimizes uplink performance. However, the gNodeB occasionally fails to obtain the strength of interference in self-contained slots. This may decrease the proportion of occasions where the controlled function can take effect when UEs continuously perform uplink full buffer services.
- After the preceding options are selected, values of certain indicators may change. Some may increase, such as **N.RLC.ThpTime.UL.Cell**, **N.ThpTime.UL**, and indicators related to the total number of PUSCH TTIs, total number of UEs scheduled in the uplink in a cell, and CCE usage. Some may decrease, such as the total number of layers that are paired for uplink MIMO, average uplink cell throughput, and number of UEs scheduled per TTI in the uplink. Moreover, some may either increase or decrease, such as uplink-PRB-related indicators, link-related indicators (such as the uplink MCS index), indicators that are used for measuring interference to symbols in self-contained slots, UE-level RSRP and SINR as well as cell-level RSSI, and the power of symbols in uplink slots. (RSRP is short for reference signal received power, SINR for signal to interference plus noise ratio, and RSSI for received signal strength indication.)
- In high-load scenarios, the downlink user-perceived rate may decrease after the function controlled by the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter takes effect. To address this issue, the PUSCH scheduling prohibition in self-contained slots function is introduced. This function is controlled by the **S_SLOT_PUSCH_SCH_PROHIB_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter. When this option is selected, the function controlled by the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter can take effect only when the uplink or downlink PRB usage is less than or equal to the value of the **NRDUCellRimConfig**.***IntrfAvoidPuschSchPrbThld*** parameter. Under this condition, the network can avoid degradation of the downlink user-perceived rate. However, the number of uplink

scheduling opportunities and the number of TTIs may decrease, which in turn reduces the uplink cell traffic volume.

**NOTE**

It is recommended that the **gNodeBAlgo**.***UlSchSwitch*** parameter be set as follows:

- Select the **8_2_SS54_ADAPT_SW** option when the single-period 8:2 slot configuration is used and the self-contained slot structure is SS54.
- Select the **8_2_SS84_ADAPT_SW** option when the dual-period 8:2 slot configuration is used and the self-contained slot structure is SS84.
- Select the **4_1_SS4_ADAPT_SW** option when the 4:1 slot configuration is used and the self-contained slot structure is SS4.

# 6.3 Requirements

## 6.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-061292 | Proactive Avoidance of Remote Interference | NR0S00PARI00 | per Cell |

For the license control item NR0S00PARI00:

- If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
- If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation:
  - When the license sales number pre-check function is enabled, the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter cannot be modified due to license insufficiency, preventing cell activation failures.
  - When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified.

The license sales number pre-check function is controlled by the **LIC_SALE_NUM_PRE_CHECK_SW** option of the **gNBLicenseAlmThld**.***LicenseChkSw*** parameter.

**NOTE**

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

## 6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 6.3.2.1 Prerequisite Functions

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| UE-characteristic-based SRS period adaptation | **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs.*SrsAlgoSwitch*** parameter | *Channel Management* | For a low-speed cell, the selection of the **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs.*SrsAlgoSwitch*** parameter is a prerequisite for the selection of the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter. |
| Scheduling optimization for HARQ-ACK resource set 0 | **HARQ_ACK_RES_SET0_SCH_OPT_SW** option of the **NRDUCellPucch**.***PucchPerformanceSw*** parameter | *Channel Management* | The selection of the **HARQ_ACK_RES_SET0_SCH_OPT_SW** option of the **NRDUCellPucch**.***PucchPerformanceSw*** parameter is a prerequisite for the selection of the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| PDCCH uplink and downlink CCE sharing | **UL_DL_CCE_SHARING_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter | *Channel Management* | The selection of the **UL_DL_CCE_SHARING_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter is a prerequisite for the selection of the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter for a cell. |
| Remote Interference Avoidance for SRS Sw and S Slot SRS Interference Coord Sw | **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter | None | The selection of the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is a prerequisite for the selection of the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter for a cell. |

### 6.3.2.2 Mutually Exclusive Functions

Functions related to UCN:

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | For a low-speed cell, the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be deselected when the **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **HYPER_CELL**. |
| Cell Combination<br>Joint Transmission Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** | *Cell Combination* | The **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be deselected if (1) the **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **HYPER_CELL_COMBINE_MODE** and (2) one of the following conditions is met:<br>● The **NRDUCell**.***LampSiteCellFlag*** parameter is set to **NO**, and the **NRDUCell**.***HighSpeedFlag*** parameter is set to **LOW_SPEED**.<br>● The **NRDUCell**.***LampSiteCellFlag*** parameter is set to **YES**, and the **NRDUCellMultiTrp**.***DataTransMode*** parameter value is neither **BASIC_MODE** nor **DPS_MODE**. |

Functions related to RAN services:

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Positioning-dedicated SRS transmission in uplink slots | **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter | *Positioning* | ● If the 2:3 slot configuration is used, the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter and the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter are mutually exclusive.<br>● When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled (by selecting the **SRS_PUSCH_CONCURRENCY_COMPT_SW** option of the **NRDUCellCarrMgmt**.***CaCompatibilitySwitch*** parameter), the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter and the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter cannot be selected together. |

Functions related to channel management:

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| SRS interference coordination based on self-contained and uplink slots | **SRS_INTRF_COORD_S_U_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter | *Channel Management* | The **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter and the **SRS_INTRF_COORD_S_U_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter are mutually exclusive. |
| CSI-RS for CM port number management | **NRDUCellCsirs**.***FR1MaxCellCsirsPortNum*** | *Channel Management* | When the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected, the **NRDUCellCsirs**.***FR1MaxCellCsirsPortNum*** parameter cannot be set to **16PORT**. |
| Short PUCCH format | **NRDUCellPucch**.***StructureType*** set to **SHORT_STRUCTURE** | *Channel Management* | When the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected, the **NRDUCellPucch**.***StructureType*** parameter must be set to **LONG_STRUCTURE**. |

Other functions:

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Master TRP of a Light DM MIMO Cell | **NRDUCellTrp**.***TrpType*** set to **MASTER_DM_MIMO_LIGHT** | None | If the **NRDUCellTrp**.***TrpType*** parameter is set to **MASTER_DM_MIMO_LIGHT**, the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be deselected. |
| CSI-RS Beam Sweeping Switch | **CSIRS_BEAM_SWEEPING_SW** option of the **NRDUCellCsirs**.***CsiSwitch*** parameter | None | The two functions are mutually exclusive: (1) proactive avoidance of remote interference (controlled by the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter) and (2) CSI-RS beam sweeping. |
| MT Access Control Switch | **MT_RRC_CONN_SWITCH** option of the **NRDUCellConnMgmt**.***IabCtrlSwitch*** parameter | None | When proactive avoidance of remote interference is enabled (by selecting the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter), the **MT_RRC_CONN_SWITCH** option of the **NRDUCellConnMgmt**.***IabCtrlSwitch*** parameter must be deselected, and the **NRDUCell**.***DeployPosition*** parameter must be set to **DEPLOY_INTEGRATED**. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Uplink DMRS Type | **NRDUCellPusch**.***UlDmrs Type*** set to **TYPE2** | None | When the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExt Switch*** parameter is selected, **NRDUCellPusch**.***UlDmrs Type*** cannot be set to **TYPE2**. |
| Common PUCCH RB resource optimization | **COMM_PUCCH_RB_RES_OPT_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter | None | The **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExt Switch*** parameter and the **COMM_PUCCH_RB_RES_OPT_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter are mutually exclusive. |
| Fallback DCI Fast Response Switch | **FALLBACK_DCI_FAST_RSP_SW** option of the **NRDUCellAlgoSwitch**.***CpLatencySwitch*** parameter | None | The **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExt Switch*** parameter and the **FALLBACK_DCI_FAST_RSP_SW** option of the **NRDUCellAlgoSwitch**.***CpLatencySwitch*** parameter are mutually exclusive. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| SRS and PUSCH/PUCCH concurrency compatibility | **SRS_PUSCH_CONCURRENCY_COMPT_SW** option of the **NRDUCellCarrMgmt**.***CaCompatibilitySwitch*** parameter | *Carrier Aggregation* | When positioning-dedicated SRS transmission in uplink slots is enabled (by selecting the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter), the SRS and PUSCH/PUCCH concurrency compatibility function and the proactive avoidance of remote interference function cannot be both enabled. |

### 6.3.2.3 Function Impacts

Functions related to carrier management:

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Flexible spectrum access (FSA) | **FLEX_SPCT_ACCESS_SW** option of the **NRDUCellCarrMgmt**.***MultiBandFlexSchSwitch*** parameter | *Uplink Flexible Spectrum Scheduling* | In 2 TDD+1 FDD three-band 3CC scenarios, if the PCC and SCC use the same slot configuration, FSA is supported only when proactive avoidance of remote interference is enabled on both the PCC and SCC.<br>If the PCC and SCC use different slot configurations, FSA is supported only when proactive avoidance of remote interference is enabled on both the PCC and SCC or only on the NR TDD cell with the single-period 8:2 slot configuration.<br>In 2 TDD+1 FDD three-band 4CC/5CC scenarios, proactive avoidance of remote interference cannot be enabled in NR TDD cells. |
| 3CC aggregation | **NRDUCellCarrMgmt**.***CaDlMaxCcNum*** set to **DL3CC** | *CA Experience Improvement* | After proactive avoidance of remote interference is enabled (by selecting the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter), more PUCCH RBs will be used in the PCell. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Massive CA | **NRDUCellCarrMgmt**.***CaDlMaxCcNum*** set to **DL4CC** or **DL5CC** | *CA Experience Improvement* | After proactive avoidance of remote interference is enabled (by selecting the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter), more PUCCH RBs will be used in the PCell. |
| Downlink intra-band CA | **INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter | *Carrier Aggregation* | After proactive avoidance of remote interference is enabled (by selecting the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter), the number of UEs on which 2CC aggregation takes effect decreases. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Downlink intra-FR inter-band CA | **INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter | *Carrier Aggregation* | After proactive avoidance of remote interference is enabled (by selecting the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter), the number of UEs on which 2CC aggregation takes effect decreases. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Uplink intra-gNodeB intra-FR inter-band CA | **INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter | *Carrier Aggregation* | ● When the switch for uplink CA is on, the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be consistently configured on the PCell and the SCell. Otherwise, differences in their uplink scheduling in self-contained slots affect the respective power headroom reporting. This results in frequent TPC adjustments, leading to significant fluctuations in the average uplink throughput of uplink-CA UEs.<br>● It is good practice to consistently configure the switch for "proactive avoidance of remote interference" on the PCell and the SCell. Otherwise, within the same symbols, SRS may be transmitted on one carrier whereas PUSCH may be transmitted on the other. This increases bit errors for some UEs. In the event of switch status |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| | | | inconsistency between the two cells, it is good practice to enable the "SRS and PUSCH/PUCCH concurrency compatibility" function to avoid bit errors. For details about the "SRS and PUSCH/PUCCH concurrency compatibility" function, see *Carrier Aggregation*. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Uplink intra-band CA | **INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter | *Carrier Aggregation* | ● When the switch for uplink CA is on, the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be consistently configured on the PCell and the SCell. Otherwise, differences in their uplink scheduling in self-contained slots affect the respective power headroom reporting. This results in frequent TPC adjustments, leading to significant fluctuations in the average uplink throughput of uplink-CA UEs.<br>● It is good practice to consistently configure the switch for "proactive avoidance of remote interference" on the PCell and the SCell. Otherwise, bit errors may increase for some UEs due to PUSCH scheduling status inconsistency between the two cells in the same slot. In the event of switch status inconsistency between the two |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| | | | cells, it is good practice to enable the "SRS and PUSCH/PUCCH concurrency compatibility" function to reduce bit errors. For details about the "SRS and PUSCH/PUCCH concurrency compatibility" function, see *Carrier Aggregation*. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| CA SRS carrier switching<br>SRS carrier switching enhancement | **SRS_CARRIER_SWITCHING_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter<br>**SRS_CARRIER_SWITCHING_ENH_SW** option of the **NRDUCellCarrMgmt.*CaEnhancedAlgoSwitch*** parameter | *CA Experience Improvement* | • In a cell with proactive avoidance of remote interference enabled, the CA SRS carrier switching switch can be turned on only if the **NRDUCellCarrMgmt**.***SrsCarrSwitchingUlSymCfg*** and **NRDUCellCarrMgmt**.***SrsCarrSwitchingTimeUlThld*** parameters are set to the same value.<br>• When proactive avoidance of remote interference is enabled, CA SRS carrier switching or CA SRS carrier switching enhancement works only in certain scenarios[a].<br>• If proactive avoidance of remote interference is enabled together with CA SRS carrier switching or CA SRS carrier switching enhancement in a cell, the following network impacts arise:<br>– CA SRS carrier switching applies less frequently to UEs that treat the cell as their |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| | | | SCell, thereby decreasing the number of UEs in the CA state in the cell.<br>– The average uplink cell and UE throughputs may decrease. |
| a: When proactive avoidance of remote interference is enabled, CA SRS carrier switching or CA SRS carrier switching enhancement works only in the following scenarios:<br>● A UE treats a TDD cell as its PCell and has one TDD SCell, the PCell and SCell use the same slot configuration, and proactive avoidance of remote interference is enabled in both cells. In this scenario, CA SRS carrier switching can take effect for the UE.<br>● A UE treats a TDD cell as its PCell and has two TDD SCells, the PCell and SCells use the same slot configuration, proactive avoidance of remote interference is enabled in the PCell, and:<br>– Proactive avoidance of remote interference is enabled in only one of the SCells. In this scenario, CA SRS carrier switching can take effect for the UE, whereas CA SRS carrier switching enhancement cannot.<br>– Proactive avoidance of remote interference is enabled in both SCells. In this scenario, CA SRS carrier switching enhancement can take effect for the UE. | | | |

Functions related to interference suppression:

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| GP symbol shutdown in special slots | **SS_GAP_SYMBOL_SHUTDN_SW** option of the **NRDUCellTimeIntrf**.***TimeIntrfAvoidSw*** parameter | *Interference Avoidance* | When the **NRDUCell**.***SlotStructure*** parameter is set to **SS54**, the following rule applies: If GP symbol shutdown in special slots is enabled and the **8_2_SS54_ADAPT_SW** option of the **gNodeBAlgo**.***UlSchSwitch*** parameter is selected, the function controlled by the **8_2_SS54_ADAPT_SW** option preferentially takes effect. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Time-domain interference avoidance | **gNodeBAlgo**.***TimeIntrfAvoidSw*** | *Interference Avoidance* | • When the **NRDUCell**.***SlotStructure*** parameter is set to **SS4**, the following rule applies: If the **4_1_SS4_ADAPT_SW** option of the **gNodeBAlgo**.***UlSchSwitch*** parameter is selected and the time-domain interference avoidance function is enabled, the time-domain interference avoidance function preferentially takes effect.<br>• When the **NRDUCell**.***SlotStructure*** parameter is set to **SS54**, the following rule applies: If the **8_2_SS54_ADAPT_SW** option of the **gNodeBAlgo**.***UlSchSwitch*** parameter is selected and the time-domain interference avoidance function is enabled, the time-domain interference avoidance function preferentially takes effect. |

<table>
<tr><th>Function Name</th><th>Function Switch</th><th>Reference</th><th>Description</th></tr>
<tr><td>Remote interference adaptive avoidance</td><td>RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter</td><td>Remote Interference Management (Low-Frequency TDD)</td><td rowspan="2">When the SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrs.SrsAlgoExtSwitch parameter are selected, the measured value of the N.UL.Last.Symbol13.Pwr counter may be greater than expected, affecting the interference detection result of remote interference management.<br><br>When the SRS_INTRF_AVOID_PUSCH_SCH_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter is selected, the measured value of the N.GAP.LastSymbol.Pwr counter may be greater than expected, affecting the interference detection result of remote interference management.</td></tr>
<tr><td>Enhanced remote interference adaptive avoidance</td><td>RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter</td><td>Remote Interference Management (Low-Frequency TDD)</td></tr>
</table>

Other functions:

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | UL and DL Decoupling does not take effect on RedCap UEs when proactive avoidance of remote interference is enabled. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| UL and DL Decoupling | **NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch*** | *UL and DL Decoupling* | |
| Uplink DMRS adaptation | **VOICE_DMRS_ADAPT_SW** option of the **NRDUCellDmrs**.***UlDmrsSwitch*** parameter | *VoNR* | When proactive avoidance of remote interference takes effect for UEs with VoNR enabled, uplink DMRS adaptation does not take effect in the uplink slots with SRS transmission. |
| Uplink fallback to LTE | **UL_FALLBACK_TO_LTE_SWITCH** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Basics* | Proactive avoidance of remote interference reduces the interference present to SRS, which lowers the uplink SINR that is considered in the determination of whether to use uplink fallback to LTE. This in turn increases the number of UEs that fall back from NR to LTE for uplink data transmissions. Accordingly, the value of **N.NsaDc.ULChangetoMeNB** increases. |
| Intelligent uplink multi-layer SINR prediction | **UL_SU_SINR_INTEL_PREDICT_SW** option of the **NRDUCellAiAlgo**.***AiAmcAlgoSwitch*** parameter | *MIMO (TDD)* | Proactive avoidance of remote interference adversely affects the proportion of occasions where intelligent uplink multi-layer SINR prediction takes effect. As a result, intelligent uplink multi-layer SINR prediction may deliver fewer benefits. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Extended Cell Range (TDD) | **NRDUCell**.***CellRadius*** set to a value greater than **14500** | *Extended Cell Range* | After proactive avoidance of remote interference is enabled (by selecting the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter), the PRACH GP is shortened, which may cause a decrease in the random access success rate of UEs on which Extended Cell Range takes effect. |
| Extended Cell Range of 90 km (TDD) | **NRDUCell**.***CellRadius*** set to a value greater than **60000** and less than or equal to **90000** | *Extended Cell Range* | After proactive avoidance of remote interference is enabled, the PRACH guard time is shortened, which may cause access failures for UEs on which Extended Cell Range of 90 km (TDD) takes effect and decrease the random access success rate. Therefore, it is not recommended that the two functions be both enabled. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Intra-base-station UL CoMP | **INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter | *CoMP* | If the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected for the serving cell but deselected for the cooperating cell, UL CoMP does not take effect. If this option is deselected for the serving cell or selected for both the serving and cooperating cells, UL CoMP can take effect. |
| Intra-base-station DL CoMP | **INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter | *CoMP* | If the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected for the serving cell but deselected for the cooperating cell, DL CoMP does not take effect. If this option is deselected for the serving cell or selected for both the serving and cooperating cells, DL CoMP can take effect. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| Co-MM | **INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter | *Co-MM (Low-Frequency TDD)* | If the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected for the serving cell but deselected for the cooperating cell, Co-MM does not take effect. If this option is deselected for the serving cell or selected for both the serving and cooperating cells, Co-MM can take effect. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| Clock out-of-synchronization detection | **CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter | *Synchronization* | After proactive avoidance of remote interference is enabled (by selecting the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter), the following impacts are introduced:<br>● If the last symbol of an uplink slot is allocated to SRS, clock out-of-synchronization detection uses SRS RBs. If the last symbol of an uplink slot is not allocated to SRS, clock out-of-synchronization detection uses unused PUSCH RBs.<br>● The measured value of the **N.UL.Last.Symbol13.Pwr** counter may be greater than expected, affecting the interference detection result of clock out-of-synchronization detection.<br>When the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected, the |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| | | | measured value of the **N.GAP.LastSymbol.Pwr** counter may be greater than expected, affecting the interference detection result of clock out-of-synchronization detection. |

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| DRX | **BASIC_DRX_SW** option of the **NRDUCellUePwrSaving.*NrDuCellDrxAlgoSwitch*** parameter | *DRX* | When proactive avoidance of remote interference (controlled by the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs.*SrsAlgoExtSwitch*** parameter) and DRX are both enabled, the DRX On Duration may be prolonged or shortened. Further details are as follows:<br>● After SRS resources are migrated to uplink slots, changes are made to the time-domain positions of the SRS and PUCCH as well as the number of PUCCH resources. As a result, the On Duration may change because it is related to the time-domain positions of the SRS and PUCCH.<br>● After SRS resources are migrated to uplink slots, interference to SRS decreases and downlink performance improves. The time required for a UE to receive the same amount of data is reduced and the UE is more likely to enter the sleep |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| | | | time. Therefore, the On Duration may be shortened. |
| Positioning-dedicated SRS transmission in uplink slots | **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter | *Positioning* | When the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter and the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter are both selected, the **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be selected. In this case, proactive avoidance of remote interference takes effect. |

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| PUSCH Sch in S Slot Sw | **PUSCH_SCH_IN_S_SLOT_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter | None | When the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter and the **PUSCH_SCH_IN_S_SLOT_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter are both selected, the **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be selected. In the case of large-packet services in the uplink, PUSCH and SRS scheduling in self-contained slots is triggered. If this occurs, proactive avoidance of remote interference does not take effect. |
| Quadruple code division for format-4 resource allocation | **FORMAT4_OCC_LENGTH_N4_SWITCH** option of the **NRDUCellPucch.*PucchAlgoSwitch*** parameter | *Channel Management* | When (1) **NRDUCell**.***HighSpeedFlag*** is set to **HIGH_SPEED** and (2) the **FORMAT4_OCC_LENGTH_N4_SWITCH** option of the **NRDUCellPucch.*PucchAlgoSwitch*** parameter is selected, enabling proactive avoidance of remote interference decreases the PUCCH resource allocation success rate. |

### 6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

#### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

#### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

#### RF Modules

All NR TDD-capable 2T2R, 4T4R, 8T8R, 32T32R, and 64T64R RF modules support this function.

### 6.3.4 Cells

The SCS is 30 kHz.

The cells work in the n38/n40/n41/n77/n78/n79 band.

The cells use the 4:1, single-period 8:2, or dual-period 8:2 slot configuration.

### 6.3.5 Others

None

## 6.4 Operation and Maintenance

### 6.4.1 Data Configuration

#### 6.4.1.1 Data Preparation

Table 6-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 6-1** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| SRS Algorithm Extension Switch | **NRDUCellSrs.*SrsAlgoExtSwitch*** | Select the **SRS_RIM_INTRF_AVOID_SW** option. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| SRS Algorithm Extension Switch | **NRDUCellSrs**.***SrsAlgoExtSwitch*** | Select the **SRS_INTRF_COORD_S_SLOT_SW** option. |
| PUCCH Receive Enhanced Switch | **NRDUCellPucch**.***PucchReceiveEnhSwitch*** | Select the **PUCCH_MRC_IRC_SW** option. |
| Schedule Multiple SLIV Type UE Switch | **NRDUCellPusch**.***SchMultiSlivTypeUeSwitch*** | It is recommended that this parameter be set to **ON**. |
| UL Scheduling Switch | **gNodeBAlgo**.***UlSchSwitch*** | ● Select the **8_2_SS54_ADAPT_SW** option when the single-period 8:2 slot configuration is used and the self-contained slot structure is SS54.<br>● Select the **8_2_SS84_ADAPT_SW** option when the dual-period 8:2 slot configuration is used and the self-contained slot structure is SS84.<br>● Select the **4_1_SS4_ADAPT_SW** option when the 4:1 slot configuration is used and the self-contained slot structure is SS4. |
| SRS Algorithm Extension Switch | **NRDUCellSrs**.***SrsAlgoExtSwitch*** | Select the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option if the single-period 8:2, dual-period 8:2, or 4:1 slot configuration is used. |
| SRS Algorithm Extension Switch | **NRDUCellSrs**.***SrsAlgoExtSwitch*** | It is recommended that the **S_SLOT_PUSCH_SCH_PROHIB_SW** option be selected if the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected. |
| Intrf Avoid PUSCH Schedule PRB Threshold | **NRDUCellRimConfig**.***IntrfAvoidPuschSchPrbThld*** | Retain the default value. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| PUSCH Interference Threshold | **NRDUCellSrs**.*PuschIntrfThld* | Retain the default value. |
| PDCCH Algorithm Switch | **NRDUCellPdcch**.*PdcchAlgoSwitch* | Select the **UL_DL_CCE_SHARING_SW** option if the single-period 8:2, dual-period 8:2, or 4:1 slot configuration is used. |

#### 6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

- Modifying the **SRS_RIM_INTRF_AVOID_SW** or **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.*SrsAlgoExtSwitch* parameter will cause automatic cell restart.
- Modifying the **8_2_SS54_ADAPT_SW**, **8_2_SS84_ADAPT_SW**, or **4_1_SS4_ADAPT_SW** option of the **gNodeBAlgo**.*UlSchSwitch* parameter on the gNodeB will result in the automatic restart of served cells with the corresponding slot configurations and self-contained slot structures.

## Activation Command Examples

```
//Turning on the switches for SRS remote interference avoidance and SRS interference coordination based on self-contained slots
MOD NRDUCELLSRS: NrDuCellId=0,
SrsAlgoExtSwitch=SRS_RIM_INTRF_AVOID_SW-1&SRS_INTRF_COORD_S_SLOT_SW-1;

//Turning on the PUCCH MRC/IRC adaptation switch
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchReceiveEnhSwitch=PUCCH_MRC_IRC_SW-1;
//Turning on the multi-SLIV UE scheduling switch
MOD NRDUCELLPUSCH: NrDuCellId=0,SchMultiSlivTypeUeSwitch=ON;
```

- Selecting the **8_2_SS54_ADAPT_SW** option when the slot configuration is single-period 8:2 and the self-contained slot structure is SS54

```
//Turning on the switch for PUSCH scheduling based on SRS interference avoidance and the switch for PUSCH scheduling prohibition in self-contained slots, and setting the PUSCH interference threshold
MOD NRDUCELLSRS: NrDuCellId=0,
SrsAlgoExtSwitch=SRS_INTRF_AVOID_PUSCH_SCH_SW-1&S_SLOT_PUSCH_SCH_PROHIB_SW-1,
PuschIntrfThld=-85;
MOD NRDUCELLRIMCONFIG: NrDuCellId=0,IntrfAvoidPuschSchPrbThld=PCT80;
```

```
//Selecting the 8_2_SS54_ADAPT_SW option
MOD GNODEBALGO:
UlSchSwitch=8_2_SS54_ADAPT_SW-1;

//Selecting the UL_DL_CCE_SHARING_SW option
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-1;
```

- Selecting the **8_2_SS84_ADAPT_SW** option when the slot configuration is dual-period 8:2 and the self-contained slot structure is SS84

```
//Turning on the switch for PUSCH scheduling based on SRS interference avoidance and the switch for PUSCH scheduling prohibition in self-contained slots, and setting the PUSCH interference threshold
MOD NRDUCELLSRS: NrDuCellId=0,
SrsAlgoExtSwitch=SRS_INTRF_AVOID_PUSCH_SCH_SW-1&S_SLOT_PUSCH_SCH_PROHIB_SW-1,
PuschIntrfThld=-85;
MOD NRDUCELLRIMCONFIG: NrDuCellId=0,IntrfAvoidPuschSchPrbThld=PCT80;
//Selecting the 8_2_SS84_ADAPT_SW option
MOD GNODEBALGO: UlSchSwitch=8_2_SS84_ADAPT_SW-1;
//Selecting the UL_DL_CCE_SHARING_SW option
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-1;
```

- Selecting the **4_1_SS4_ADAPT_SW** option when the slot configuration is 4:1 and the self-contained slot structure is SS4

```
//Turning on the switch for PUSCH scheduling based on SRS interference avoidance and the switch for PUSCH scheduling prohibition in self-contained slots, and setting the PUSCH interference threshold
MOD NRDUCELLSRS: NrDuCellId=0,
SrsAlgoExtSwitch=SRS_INTRF_AVOID_PUSCH_SCH_SW-1&S_SLOT_PUSCH_SCH_PROHIB_SW-1,
PuschIntrfThld=-85;
MOD NRDUCELLRIMCONFIG: NrDuCellId=0,IntrfAvoidPuschSchPrbThld=PCT80;
//Selecting the 4_1_SS4_ADAPT_SW option
MOD GNODEBALGO: UlSchSwitch=4_1_SS4_ADAPT_SW-1;
//Selecting the UL_DL_CCE_SHARING_SW option
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-1;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Deselecting the **8_2_SS54_ADAPT_SW** option when the slot configuration is single-period 8:2 and the self-contained slot structure is SS54

```
//Deselecting the 8_2_SS54_ADAPT_SW option
MOD GNODEBALGO: UlSchSwitch=8_2_SS54_ADAPT_SW-0;
//Turning off the switch for PUSCH scheduling based on SRS interference avoidance and the switch for PUSCH scheduling prohibition in self-contained slots
MOD NRDUCELLSRS: NrDuCellId=0,
SrsAlgoExtSwitch=SRS_INTRF_AVOID_PUSCH_SCH_SW-0&S_SLOT_PUSCH_SCH_PROHIB_SW-0;
//Deselecting the UL_DL_CCE_SHARING_SW option
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-0;
```

- Deselecting the **8_2_SS84_ADAPT_SW** option when the slot configuration is dual-period 8:2 and the self-contained slot structure is SS84

```
//Deselecting the 8_2_SS84_ADAPT_SW option
MOD GNODEBALGO: UlSchSwitch=8_2_SS84_ADAPT_SW-0;
//Turning off the switch for PUSCH scheduling based on SRS interference avoidance and the switch for PUSCH scheduling prohibition in self-contained slots
MOD NRDUCELLSRS: NrDuCellId=0,
SrsAlgoExtSwitch=SRS_INTRF_AVOID_PUSCH_SCH_SW-0&S_SLOT_PUSCH_SCH_PROHIB_SW-0;
//Deselecting the UL_DL_CCE_SHARING_SW option
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-0;
```

- Deselecting the **4_1_SS4_ADAPT_SW** option when the slot configuration is 4:1 and the self-contained slot structure is SS4

```
//Deselecting the 4_1_SS4_ADAPT_SW option
MOD GNODEBALGO: UlSchSwitch=4_1_SS4_ADAPT_SW-0;
//Turning off the switch for PUSCH scheduling based on SRS interference avoidance and the switch for PUSCH scheduling prohibition in self-contained slots
```

```
MOD NRDUCELLSRS: NrDuCellId=0,
SrsAlgoExtSwitch=SRS_INTRF_AVOID_PUSCH_SCH_SW-0&S_SLOT_PUSCH_SCH_PROHIB_SW-0;
//Deselecting the UL_DL_CCE_SHARING_SW option
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-0;
```

```
//Deselecting the SRS_INTRF_AVOID_PUSCH_SCH_SW option
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SRS_INTRF_AVOID_PUSCH_SCH_SW-0;
//Turning off the SRS remote interference avoidance switch
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SRS_RIM_INTRF_AVOID_SW-0;
//Turning off the PUCCH MRC/IRC adaptation switch
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchReceiveEnhSwitch=PUCCH_MRC_IRC_SW-0;
//Turning off the multi-SLIV UE scheduling switch
MOD NRDUCELLPUSCH: NrDuCellId=0,SchMultiSlivTypeUeSwitch=OFF;
```

### 6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 6.4.2 Activation Verification

1. Start Uu signaling tracing: Log in to the MAE-Access and choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. On the displayed page, choose **Trace Type** > **NR** > **Application Layer** > **Uu Interface Trace**.
2. View the RRCReconfiguration message tracing result. Observe the SRS time-domain position information in the *SRS-Resource* IE in SRS-Config of the RRC signaling. If the SRS time-domain position is in an uplink slot, proactive avoidance of remote interference has taken effect. **Figure 6-3** shows the *SRS-Resource* IE, in which **startPosition** indicates the SRS symbol position and **periodicityAndOffset-p** indicates the SRS slot position.

**Figure 6-3** SRS-Resource IE

```
SRS-Resource ::=                        SEQUENCE {
    srs-ResourceId                          SRS-ResourceId,
    nrofSRS-Ports                           ENUMERATED {port1, ports2, ports4},
    ptrs-PortIndex                          ENUMERATED {n0, n1 }
    transmissionComb                        CHOICE {
        n2                                      SEQUENCE {
            combOffset-n2                           INTEGER (0..1),
            cyclicShift-n2                          INTEGER (0..7)
        },
        n4                                      SEQUENCE {
            combOffset-n4                           INTEGER (0..3),
            cyclicShift-n4                          INTEGER (0..11)
        }
    },
    resourceMapping                         SEQUENCE {
        startPosition                           INTEGER (0..5),
        nrofSymbols                             ENUMERATED {n1, n2, n4},
        repetitionFactor                        ENUMERATED {n1, n2, n4}
    },
    freqDomainPosition                      INTEGER (0..67),
    freqDomainShift                         INTEGER (0..268),
    freqHopping                             SEQUENCE {
        c-SRS                                   INTEGER (0..63),
        b-SRS                                   INTEGER (0..3),
        b-hop                                   INTEGER (0..3)
    },
    groupOrSequenceHopping                  ENUMERATED { neither, groupHopping, sequ
    resourceType                            CHOICE {
        aperiodic                               SEQUENCE {
            ...
        },
        semi-persistent                         SEQUENCE {
            periodicityAndOffset-sp                     SRS-PeriodicityAndOffset,
            ...
        },
        periodic                                SEQUENCE {
            periodicityAndOffset-p                      SRS-PeriodicityAndOffset,
            ...
        }
```

## 6.4.3 Network Monitoring

Observe whether related indicators change as expected by referring to **6.2.1 Benefits** and **6.2.2 Impacts**.

# 7 Symbol Configuration for Channel Calibration

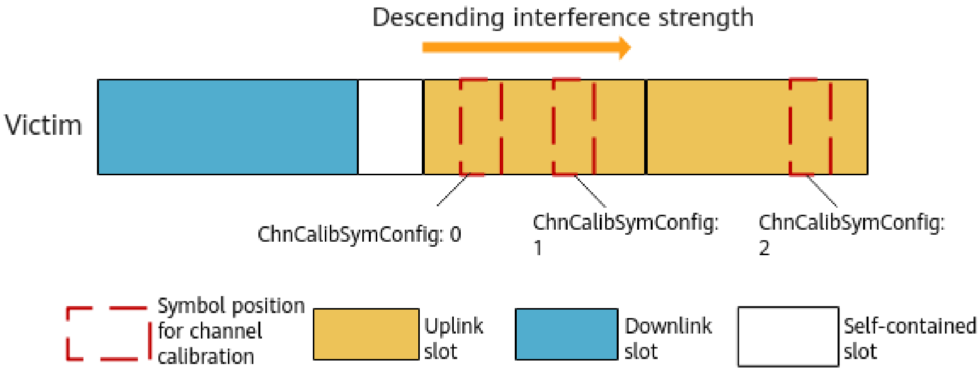
## 7.1 Principles

If remote interference is present in uplink slots, channel calibration fails. To address this issue, symbol configuration for channel calibration is introduced.

This function enables channel calibration sequences to be transmitted in an uplink slot's symbols that suffer from weak remote interference, as shown in **Figure 7-1**. This mitigates the impacts of remote interference on channel calibration, thereby increasing the channel calibration success rate. The **NRDUCellChnCalib**.***ChnCalibSymConfig*** parameter determines the positions of symbols for channel calibration sequences, as described below:

- If this parameter is set to **0**, the positions of symbols for channel calibration sequences remain unchanged.
- If this parameter is set to **1**, channel calibration sequences are transmitted in the later symbols of the baseline uplink slot.
- If this parameter is set to **2**, channel calibration sequences are transmitted in the next uplink slot following the baseline uplink slot.

**Figure 7-1** Symbol configuration for channel calibration



# 7.2 Network Analysis

## 7.2.1 Benefits

It is recommended that this function be enabled if channel calibration fails due to remote interference. When this function is enabled (with the **NRDUCellChnCalib**.***ChnCalibSymConfig*** parameter set to **1** or **2**), it lessens the impact of remote interference on channel calibration, thereby increasing the channel calibration success rate. This in turn increases the **Cell Downlink Average Throughput (DU)**.

In case of strong interference, it is recommended that proactive avoidance of remote interference also be enabled. For details, see **6 Proactive Avoidance of Remote Interference**.

## 7.2.2 Impacts

The impacts between this function and other functions are described in **7.3.2.3 Function Impacts**.

There is no impact on the network.

# 7.3 Requirements

## 7.3.1 Licenses

There are no license requirements for basic functions.

## 7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

### 7.3.2.1 Prerequisite Functions

None

### 7.3.2.2 Mutually Exclusive Functions

None

### 7.3.2.3 Function Impacts

None

### 7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

#### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

#### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

#### RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

### 7.3.4 Cells

If the **NRDUCell**.***SlotAssignment*** parameter is set to **4_1_DDDSU** or **1_1_DSDSDSDSDS**, the **NRDUCellChnCalib**.***ChnCalibSymConfig*** parameter cannot be set to **2**.

### 7.3.5 Others

None

## 7.4 Operation and Maintenance

### 7.4.1 Data Configuration

#### 7.4.1.1 Data Preparation

**Table 7-1** describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 7-1** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| Channel Calibration Symbol Configuration | **NRDUCellChnCalib**.***ChnCalibSymConfig*** | If channel calibration fails due to remote interference, it is recommended that this parameter be set to **1** or **2**. |

#### 7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
MOD NRDUCELLCHNCALIB: NrDuCellId=0, ChnCalibSymConfig=1;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
MOD NRDUCELLCHNCALIB: NrDuCellId=0, ChnCalibSymConfig=0;
```

#### 7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 7.4.2 Activation Verification

This function has taken effect if the value of **Calibration Result** has changed from any other value to **Succeeded** in the **DSP NRDUCELLCHNCALIB** command output.

### 7.4.3 Network Monitoring

Observe whether related indicators change as expected by referring to **7.2.1 Benefits**.

# 8 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

**NOTE**

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

**FAQ 1: How do I find the parameters related to a certain feature from parameter reference?**

**Step 1** Open the EXCEL file of parameter reference.

**Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

**Step 3** Click **OK**. All parameters related to the feature are displayed.

**----End**

**FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?**

**Step 1** Open the EXCEL file of the used reserved parameter list.

**Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

**----End**

# 9 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

**NOTE**

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

**FAQ: How do I find the counters related to a certain feature from performance counter reference?**

**Step 1** Open the EXCEL file of performance counter reference.

**Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

**Step 3** Click **OK**. All counters related to the feature are displayed.

**----End**

# 10 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 11 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *CoMP*
- *Beam Management*
- *Hyper Cell*
- *Cell Combination*
- *High Speed Mobility*
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*
- *Synchronization*
- 3GPP TS 38.211: "NR; Physical channels and modulation"
- 3GPP TS 28.541: "Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3"
- *Channel Management*
- *Carrier Aggregation*
- *Indoor Coverage*
- *Interference Avoidance*
- *Energy Conservation and Emission Reduction*
- *Standards Compliance*
- *Extended Cell Range*
- *Virtual Grid-based Multi-Frequency Coordination*
- *Cell Management*
- *DRX*
- *Positioning*
- *5G Networking and Signaling*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Multi-RAT Coordinated Energy Saving*
- *License Control Item Lists*
- *Green Site*
- *CA Experience Improvement*

- *MIMO (TDD)*
- *Turbo Uplink (Low-Frequency TDD)*
- *EPC-based NSA Basics*
- *Multi-Frequency Smart Aggregation*
- *Uplink Flexible Spectrum Scheduling*
- *RedCap*
- *VoNR*
- *Co-MM (Low-Frequency TDD)*
- *UL and DL Decoupling*